exp_switchback_carryover_adaptation

Specialization: unknown_decay_adaptation_index

TL;DR. In a one-unit, fixed-array switchback experiment with assignment as the only randomness, the stationary Markov washout rule at scale b has exact worst-case auxiliary homogeneous-history variance $B^2V^{\dagger}_{T,b}$, sharp uniform coefficient $8e-4$, and exposure floor with unattained infimum $1/(2e)$. For $t \geq b+2$, $b+1 \leq \ell \leq t-1$, and $a \in \{0,1\}$, reversibility gives $\Pr_{\pi^{(b)}}(W_{t-\ell} \neq a \mid H_{t,b,a} = 1) = (1-\lambda_b^{\ell-b})/2$, yielding the computable replacement certificate $E^{\mathrm{cond}}_{T,\beta}(b)$, risk bound $U^{\mathrm{cond}}_{T,\beta}(b)$, and oracle ceiling exponent $\beta/(2\beta+1)$. A second fixed-before-assignment branch draws one fair coin and holds that arm for the entire horizon; its exact worst-case risk is B. It sharpens both beta-aware and common finite selectors from a $2B$ elbow to a B elbow. The exact identities $R_1(\beta; C) = B$ and $A_1(C) = 1$ persist. For growing horizons, the exact oracle order and the unknown-decay adaptation dichotomy remain open.

1 Project justification

Gap. Known-order switchback results do not pose the min-over-designs, max-over-the-pair (Y, β) problem over all sequential outcome-adaptive kernels. The supported priors give policy-uniform lower obstructions within individual classes but not the two-class cross-law tradeoff required for divergent adaptation. The exact B-risk persistent branch tightens the finite numerator selector, yet no common $T^{o(1)}$ multiscale procedure, matching oracle lower rate, or sharp order of $A_T(C)$ is known.

Niche. The note indexes nested fixed-array historical-envelope classes directly by an unknown polynomial decay exponent, without a distributed-lag equation, outcome noise, or a superpopulation, and permits fully sequential assignment. It supplies evaluable finite design guarantees, including an exact B-risk persistent homogeneous-path selector branch, and isolates why within-class lower obstructions do not yet yield an across-class adaptation theorem.

Fill. For every washout scale b, the note proves the exact worst-case auxiliary homogeneous-history variance $B^2V^{\dagger}_{T,b}$, its sharp uniform coefficient $8e-4$, and the exact conditional pre-window mismatch identity on the domain $t \geq b+2$, $b+1 \leq \ell \leq t-1$, $a \in \{0,1\}$. These yield $U^{\mathrm{cond}}_{T,\beta}(b)$ and the beta-aware ceiling coefficient $\sqrt{(8e-4)(1+2^{-1/(2\beta+1)})}\,B + C/\beta$. In addition, the persistent homogeneous-path fair-coin procedure has fixed-array risk $(|\mu_0| + |\mu_1|)/2 \leq B$ and exact worst-case risk B, so finite beta-aware and common selectors use a B elbow. The exact binomial oracle-risk floor, the Markov exposure floor, and the one-period identities remain unchanged. The K1/K2 dichotomy is still the disclosed residual.

2 Related work

Bojinov, Simchi-Levi, and Zhao study switchback experiments under a known finite carryover order, and Yu, Ma, and Liu optimize under a known carryover specification. Zeng and coauthors allow outcome-history-adaptive sequential rerandomization under no or first-order carryover; Xiong, Chin, and Taylor analyze modeled bias–variance tradeoffs; Hu and Wager use geometric mixing; and Glynn, Johari, and Rasouli study adaptive reusable-state Markov models. None poses one common design-estimator pair against the nested pair (Y, β) in the fixed-array historical-envelope model. The present theorem's finite selector now includes an exact B-risk persistent homogeneous-path fair-coin branch, but this finite improvement does not resolve the unknown-decay adaptation problem.

The closest accepted-bank experimental-design minimax result is `exp_rollout_chebyshev_minimax_tv_envelope_rollout_des`
It chooses treated-fraction nodes for a many-unit network rollout whose mean is a polynomial and proves a matched Chebyshev–Lobatto frontier under a total-variation variance envelope. That theorem does not subsume the present one-unit result: it has no temporal potential-outcome history, Markov cadence, carryover replacement error, fully sequential policy class, or unknown decay index. Conversely, the present theorem does not optimize rollout-node geometry, use polynomial extrapolation, or prove a matched network-rollout minimax frontier.

3 Setup

Target (estimand / structural parameter): $\mathrm{GATE}_T(Y) = T^{-1}\sum_{t=1}^{T}[Y_t(\mathbf{1}_{1:t}) - Y_t(\mathbf{0}_{1:t})]$. Primary functional / estimator / diagnostic: For $C > 0$, let

$$C_0^\star = \sqrt{(8e-4)\left(1+2^{-1/(2\beta_+ +1)}\right)}\,B + \frac{C}{\beta_-}.$$

For $b \in \{1,\dots,T\}$, put $r_b = b/(b+1)$, $k_t = \min\{b, t-1\}$, $q_b = (1+1/b)^b$, and $\lambda_b = (b-1)/(b+1)$, and define

$$V_{T,b}^\dagger = 4\sum_{t=1}^{T} r_b^{-k_t} + 8\sum_{h=1}^{b}\sum_{t=1}^{T-h} r_b^{h-k_{t+h}} + 8\sum_{d=1}^{(T-b-1)_+}(T-b-d)\lambda_b^d,$$

$$E_{T,\beta}^{\mathrm{cond}}(b) = \frac{2}{T}\sum_{t=b+2}^{T}\min\left\{2B, \frac{C}{2}\sum_{\ell=b+1}^{t-1}(1+\ell)^{-(1+\beta)}(1-\lambda_b^{\ell-b})\right\}, \qquad U_{T,\beta}^{\mathrm{cond}}(b) = \frac{B}{T}\sqrt{V_{T,b}^\dagger} + E_{T,\beta}^{\mathrm{cond}}(b),$$

with empty inner sums equal to zero. The Markov branch at scale b has

$$\upsilon_{t,b,a} = \frac{1}{2}r_b^{k_t} \geq \frac{1}{2q_b} > \frac{1}{2e},$$

and the unattained uniform infimum over admissible T, b, t, a is $1/(2e)$. For $t \geq b+2$, $b+1 \leq \ell \leq t-1$, and $a \in \{0,1\}$,

$$\Pr_{\pi^{(b)}}(W_{t-\ell} \neq a \mid H_{t,b,a} = 1) = \frac{1-\lambda_b^{\ell-b}}{2}.$$

Its exact worst-case auxiliary homogeneous-history variance is $B^2 V_{T,b}^\dagger$, and

$$V_{T,b}^\dagger \leq 4q_bT + 8Tb(q_b-1) + 4(T-b)(b-1) < (8e-4)Tb,$$

with

$$\lim_{b\to\infty}\lim_{T\to\infty}\frac{V_{T,b}^\dagger}{Tb} = 8e-4.$$

Thus $8e-4$ is the optimal uniform coefficient for this auxiliary variance inequality. If $b_\beta = \lceil T^{1/(2\beta+1)}\rceil$, define

$$\mathfrak{U}_{T,\beta}^{\mathrm{cond}} = \begin{cases} B, & T = 1, \\ \min\left\{B, \min_{1\leq b\leq T} U_{T,\beta}^{\mathrm{cond}}(b)\right\}, & T \geq 2. \end{cases}$$

The competing B branch draws $R \sim \mathrm{Bernoulli}(1/2)$ before outcomes, sets $W_t = R$ for every t, and uses

$$\widehat{\theta}^{\mathrm{hom}} = (2R-1)T^{-1}\sum_{t=1}^{T} Y_t^{\mathrm{obs}}.$$

For $\mu_a = T^{-1}\sum_{t=1}^{T} Y_t(\mathbf{a}_{1:t})$, its fixed-array risk is $(|\mu_0| + |\mu_1|)/2 \leq B$; the constant admissible array $Y_t(\cdot) = B$ attains equality, so its exact worst-case risk is B. The beta-aware finite selector chooses before assignment between this persistent branch and the smallest minimizing Markov scale. Then

$$\frac{B}{\sqrt{2T}} \leq B2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2\rfloor} \leq R_T(\beta;C) \leq \mathfrak{U}_{T,\beta}^{\mathrm{cond}} \leq \begin{cases} B, & T=1, \\ U_{T,\beta}^{\mathrm{cond}}(b_\beta), & T\geq 2, \end{cases} \leq \left[\sqrt{(8e-4)\left(1+2^{-1/(2\beta+1)}\right)}\,B + \frac{C}{\beta}\right]T^{-\beta/(2\beta+1)} \leq C_0^\star T^{-\beta/(2\beta+1)}.$$

At $T=1$, $R_1(\beta;C)=B$, supplied by the explicit fair-Bernoulli one-period procedure rather than by $U^{\mathrm{cond}}_{1,\beta}(1)$. For $b_-=\lceil T^{1/(2\beta_-+1)}\rceil$, define

$$N_T^{\mathrm{fin}}=\begin{cases}B, & T=1,\\ U^{\mathrm{cond}}_{T,\beta_-}(b_-), & T\ge 2,\end{cases}\qquad N_T^\star=\begin{cases}B, & T=1,\\ \min\left\{B,\min_{1\le b\le T}U^{\mathrm{cond}}_{T,\beta_-}(b)\right\}, & T\ge 2,\end{cases}$$

The B branch in $N_T^\star$ is the same exact-risk persistent homogeneous-path fair-coin procedure. The alternative Markov branch is tuned only to the known endpoint β_-, so the common selector is independent of the realized β. Define

$$D_T^{\mathrm{bin}}=B2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2\rfloor}.$$

Then

$$1\le A_T(C)\le \frac{N_T^\star}{D_T^{\mathrm{bin}}}\le\frac{N_T^{\mathrm{fin}}}{D_T^{\mathrm{bin}}}\le\frac{\sqrt{2}}{B}\left[\sqrt{(8e-4)\left(1+2^{-1/(2\beta_-+1)}\right)}\,B+\frac{C}{\beta_-}\right]T^{1/(4\beta_-+2)}.$$

At $T=1$, $N_1^\star=N_1^{\mathrm{fin}}=D_1^{\mathrm{bin}}=B$ and $A_1(C)=1$. The auxiliary variance exactness is not exact estimator-risk, bias, carryover-error, or oracle-risk exactness. The conditional carryover term is an expected replacement-error upper bound, not an exact bias or maximal error. The adaptation ratios use the exact binomial Bayes lower bound as an oracle-risk denominator; they do not use the unknown exact oracle risk except at $T=1$. The last coefficient is endpoint-specific and is not replaced by the larger β_+-uniform constant $C_0^\star$.

- `T` — `integer`; `\mathbb N`; `asymptotic index`

  Definition. `experiment horizon`

- `B` — `scalar`; `(0,\infty)`; `regularity constant`

  Definition. `uniform potential-outcome bound`

- `C` — `scalar`; `[0,\infty)`; `regularity constant`

  Definition. `historical carryover radius`

- `\beta_-` — `scalar`; `(0,\infty)`; `adaptation-range endpoint`

  Definition. `lower endpoint of the decay range`

- `\beta_+` — `scalar`; `(\beta_-,\infty)`; `adaptation-range endpoint`

  Definition. `upper endpoint of the decay range`

- `\beta` — `scalar`; `[\beta_-,\beta_+]`; `smoothness index`

  Definition. `polynomial historical-decay index`

- `m` — `integer`; `\{0,\ldots,T-2\}`; `comparator index`

  Definition. `finite carryover order`

- `b` — `integer`; `\{1,\ldots,T\}`; `design scale`

  Definition. `washout scale`

- `W` — `random vector`; `\{0,1\}^T`; `only experimental randomness`

  Definition. `binary assignment path`

- `Y` — `array`; `Y_t:\{0,1\}^t\to[-B,B]\text{ for }t=1,\ldots,T`; `finite population`

  Definition. `fixed potential-outcome array`

- `Y^{\mathrm{obs}}` — `vector`; `[-B,B]^T`; `observed data`

  Definition. observed outcomes, with $Y_t^{\mathrm{obs}} = Y_t(W_{1:t})$

- `\pi` — `probability kernel`; `\pi_t(\cdot\mid W_{1:t-1},Y_{1:t-1}^{\mathrm{obs}})`; `oracle design decision`

  Definition. `possibly outcome-adaptive sequential assignment rule`

- `\widehat\theta` — `estimator`; `\widehat\theta(W,Y^{\mathrm{obs}})`; `analysis decision`

  Definition. `data-measurable GATE estimator`

- `\operatorname{GATE}_T` — `functional`; `Y\mapsto T^{-1}\sum_{t=1}^T[Y_t(\mathbf 1_{1:t})-Y_t(\mathbf 0_{1:t})]`; `causal estimand`

  Definition. `all-treated versus all-control global average treatment effect`

- `w^{(\beta)}` — `sequence`; `w^{(\beta)}_\ell=C(1+\ell)^{-(1+\beta)}`; `carryover modulus`

  Definition. `historical impulse-response envelope`

- `\mathcal P_{T,\beta}(C)` — `set`; `nested adversarial class`

  Definition. `bounded polynomial historical-envelope class`

- `\mathcal P_T^{(0)}` — `set`; `sanity-check class`

  Definition. `no-carryover class with unrestricted direct effects`

- `\mathcal F_{T,m}` — `set`; `scope comparator`

  Definition. `finite-order carryover class`

- `\mathcal D_T` — `set`; `oracle design class`

  Definition. `all sequential assignment kernels`

- `R_T(\beta;C)` — `scalar functional`; `[0,\infty)`; `oracle benchmark`

  Definition. `decay-aware minimax absolute risk`

- `R_T^{(0)}` — `scalar functional`; `[0,\infty)`; `sanity benchmark`

  Definition. `no-carryover minimax absolute risk`

- `A_T(C)` — `scalar functional`; `[1,\infty]`; `focal object`

  Definition. `unknown-decay adaptation index`

- `a_m` — `scalar`; `(0,\infty)`; `finite-order scope witness`

  Definition. lag-$m+1$ witness amplitude when $C > 0$

- `\gamma_\ell` — `sequence`; `[0,\infty)`; `finite-order scope witness`

  Definition. `lag coefficients for an infinite-tail scope witness`

- `g_T` — `scalar functional`; `[1,\infty]`; `open adaptation-divergence factor`

  Definition. `strongest two-class adaptation lower-bound factor delivered by the open policy-uniform cadence-dichotomy construction`

- `C_0^\star` — `scalar`; `(0,\infty)`; `theorem constant`

  Definition. uniform sharpened oracle upper-rate constant $C_0^\star = \sqrt{(8e-4)(1+2^{-1/(2\beta_+ +1)})}\,B + C/\beta_-$

4 Assumptions

Assumption 1 (ass:bounded-array)**.** *$|Y_t(z_{1:t})| \le B$ for every $t \in \{1,\dots,T\}$ and $z_{1:t} \in \{0,1\}^t$.*

Novel. *After an operational outcome scale is fixed, bounded platform metrics are natural finite-population restrictions and permit design-based worst-case risk control.*

Assumption 2 (ass:historical-polynomial-envelope)**.** *For every $t, z_{1:t}, z'_{1:t}$ with $z_t = z'_t$,*

$$|Y_t(z_{1:t}) - Y_t(z'_{1:t})| \le \sum_{\ell=1}^{t-1} C(1+\ell)^{-(1+\beta)} \mathbf{1}\{z_{t-\ell} \ne z'_{t-\ell}\}$$

.

Novel. *A pathwise polynomial historical modulus represents persistent queues, inventories, and marketplace states without a distributed-lag equation; it deliberately places no restriction on the contemporaneous treatment contrast.*

Assumption 3 (ass:no-carryover)**.** *$Y_t(z_{1:t}) = Y_t(z'_{1:t})$ whenever $z_t = z'_t$.*

Standard *(zero carryover, [1]).*

Assumption 4 (ass:finite-order-carryover)**.** *$Y_t(z_{1:t}) = Y_t(z'_{1:t})$ whenever $z_{(t-m)\vee 1:t} = z'_{(t-m)\vee 1:t}$.*

Standard *(finite carryover order, [1]).*

5 Definitions

Definition 1 (def:decay-weights)**.** Defined object. $w^{(\beta)}$

Construction. *$w^{(\beta)}_\ell = C(1+\ell)^{-(1+\beta)}$ for $\ell = 1,\dots,T-1$.*

Definition 2 (def:poly-class)**.** Defined object. $\mathcal{P}_{T,\beta}(C)$

Construction.

$$\mathcal{P}_{T,\beta}(C) = \{Y : |Y_t(z_{1:t})| \le B \text{ for all } t, z_{1:t},\ \text{and } |Y_t(z_{1:t}) - Y_t(z'_{1:t})| \le \sum_{\ell=1}^{t-1} C(1+\ell)^{-(1+\beta)} \mathbf{1}\{z_{t-\ell} \ne z'_{t-\ell}\} \text{ whenever } z_t = z'_t,\ 1 \le t \le T\}$$

.

Member properties. *`ass:bounded-array`; `ass:historical-polynomial-envelope`.*

Definition 3 (def:no-carryover-class)**.** Defined object. $\mathcal{P}^{(0)}_T$

Construction.

$$\mathcal{P}^{(0)}_T = \{Y : |Y_t(z_{1:t})| \le B \text{ for all } t, z_{1:t},\ Y_t(z_{1:t}) = Y_t(z'_{1:t}) \text{ whenever } z_t = z'_t,\ 1 \le t \le T\}$$

.

Member properties. *`ass:bounded-array`; `ass:no-carryover`.*

Definition 4 (def:finite-order-class)**.** Defined object. $\mathcal{F}_{T,m}$

Construction.

$$\mathcal{F}_{T,m} = \{Y : |Y_t(z_{1:t})| \le B \text{ for all } t, z_{1:t},\ Y_t(z_{1:t}) = Y_t(z'_{1:t}) \text{ whenever } z_{(t-m)\vee 1:t} = z'_{(t-m)\vee 1:t},\ 1 \le t \le T\}$$

.

Member properties. *`ass:bounded-array`; `ass:finite-order-carryover`.*

Definition 5 (def:design-class)**.** Defined object. $\mathcal{D}_T$

Construction.

$$\mathcal{D}_T = \{\pi : \pi_t(\cdot \mid W_{1:t-1}, Y^{\text{obs}}_{1:t-1}) \text{ is a probability law on } \{0,1\},\ t = 1,\dots,T\}$$

.

Definition 6 (def:gate)**.** Defined object. GATE_T

Construction. $\text{GATE}_T(Y) = T^{-1} \sum_{t=1}^T [Y_t(\mathbf{1}_{1:t}) - Y_t(\mathbf{0}_{1:t})]$.

Definition 7 (def:oracle-risk)**.** Defined object. $R_T(\beta; C)$
Construction.

$$R_T(\beta; C) = \inf_{\pi\in\mathcal{D}_T} \inf_{\widehat{\theta}} \sup_{Y\in\mathcal{P}_{T,\beta}(C)} \mathbb{E}_\pi |\widehat{\theta} - \mathrm{GATE}_T(Y)|$$

.

Definition 8 (def:no-carryover-risk)**.** Defined object. $R_T^{(0)}$
Construction.

$$R_T^{(0)} = \inf_{\pi\in\mathcal{D}_T} \inf_{\widehat{\theta}} \sup_{Y\in\mathcal{P}_T^{(0)}} \mathbb{E}_\pi |\widehat{\theta} - \mathrm{GATE}_T(Y)|$$

.

Definition 9 (def:adaptation-index)**.** Defined object. $A_T(C)$
Construction.

$$A_T(C) = \inf_{\pi\in\mathcal{D}_T} \inf_{\widehat{\theta}} \sup_{\beta\in[\beta_-,\beta_+]} \frac{\sup_{Y\in\mathcal{P}_{T,\beta}(C)} \mathbb{E}_\pi |\widehat{\theta} - \mathrm{GATE}_T(Y)|}{R_T(\beta; C)}$$

.

6 Main results

Theorem 1 (thm:oracle-rate)**.** *Fix $C > 0$ and set*

$$C_0^\star = \sqrt{(8e-4)\left(1 + 2^{-1/(2\beta_+ + 1)}\right)}\, B + \frac{C}{\beta_-}.$$

Uniformly over $\beta \in [\beta_-, \beta_+]$,

$$\frac{B}{\sqrt{2T}} \le B 2^{-(T-1)} \binom{T-1}{\lfloor (T-1)/2 \rfloor} \le R_T(\beta; C).$$

At $T = 1$,

$$R_1(\beta; C) = B,$$

attained from above by $W_1 \sim \mathrm{Bernoulli}(1/2)$ and $\widehat{\theta}_1 = (2W_1 - 1)Y_1^{\mathrm{obs}}$. For $b \in \{1, \dots, T\}$, put $r_b = b/(b+1)$, $k_t = \min\{b, t-1\}$, $\lambda_b = (b-1)/(b+1)$,

$$V_{T,b}^\dagger = 4\sum_{t=1}^{T} r_b^{-k_t} + 8\sum_{h=1}^{b}\sum_{t=1}^{T-h} r_b^{h-k_{t+h}} + 8\sum_{d=1}^{(T-b-1)_+} (T-b-d)\lambda_b^d,$$

$$E_{T,\beta}^{\mathrm{cond}}(b) = \frac{2}{T}\sum_{t=b+2}^{T} \min\left\{2B, \frac{C}{2}\sum_{\ell=b+1}^{t-1} (1+\ell)^{-(1+\beta)}(1-\lambda_b^{\ell-b})\right\}, \qquad U_{T,\beta}^{\mathrm{cond}}(b) = \frac{B}{T}\sqrt{V_{T,b}^\dagger} + E_{T,\beta}^{\mathrm{cond}}(b),$$

with empty inner sums equal to zero. Define the integrated finite oracle certificate

$$\mathfrak{U}_{T,\beta}^{\mathrm{cond}} = \begin{cases} B, & T = 1, \\ \min\left\{B, \min_{1\le b\le T} U_{T,\beta}^{\mathrm{cond}}(b)\right\}, & T \ge 2. \end{cases}$$

Put $b_\beta = \lceil T^{1/(2\beta+1)} \rceil$*. Then*

$$R_T(\beta; C) \le \mathfrak{U}^{\mathrm{cond}}_{T,\beta} \le \begin{cases} B, & T=1, \\ U^{\mathrm{cond}}_{T,\beta}(b_\beta), & T \ge 2, \end{cases} \le \left[\sqrt{(8e-4)\left(1+2^{-1/(2\beta+1)}\right)}\, B + \frac{C}{\beta} \right] T^{-\beta/(2\beta+1)} \le C_0^\star T^{-\beta/(2\beta+1)}.$$

For $T \ge 2$*, each Markov branch is the explicit beta-aware stationary Markov switchback and observed-data Horvitz–Thompson estimator of Lemma* $\mathrm{lem:markov-washout-oracle-upper}$*. The alternative* B *branch draws* $R \sim \mathrm{Bernoulli}(1/2)$ *before outcomes, sets* $W_t = R$ *for every* t*, and uses*

$$\widehat{\theta}^{\mathrm{hom}} = (2R-1)T^{-1} \sum_{t=1}^{T} Y_t^{\mathrm{obs}}.$$

Writing $\mu_a = T^{-1}\sum_{t=1}^{T} Y_t(\mathbf{a}_{1:t})$*, its fixed-array risk is* $(|\mu_0|+|\mu_1|)/2 \le B$*, and its exact worst-case risk is* B*, attained by* $Y_t(z_{1:t}) = B$*. The finite minimum is attained, and one may select before assignment between the persistent homogeneous-path branch and the smallest minimizing Markov scale using only* T, B, C, β*. The* $T=1$ *value* B *comes from the explicit one-period procedure, not from an assertion that* $U^{\mathrm{cond}}_{1,\beta}(1) = B$*. The lower bound holds for every* $\pi \in \mathcal{D}_T$*, including deterministic, history-dependent, and outcome-adaptive kernels without a positivity restriction. The Markov branches have* $\upsilon_{t,b,a} \ge 1/(2q_b) > 1/(2e)$*, where* $q_b = (1+1/b)^b$*, with unattained uniform infimum* $1/(2e)$*. Their auxiliary homogeneous-history variance has exact worst-case value* $B^2 V^\dagger_{T,b}$*, whose sharp uniform coefficient is* $8e-4$*. Their carryover terms are conditional, endpoint-aware expected replacement-error upper bounds obtained from the exact exposure-weight identity and the conditional mismatch formula, not exact biases or maximal-bias formulas. The auxiliary variance identity is not an exact estimator-risk identity. The lower and upper exponents differ; no matching lower bound at the upper exponent and no equality between the optimized certificate and oracle risk is claimed.*

Proof. **Proof.** Lemma $\mathrm{lem:positive-oracle-risk}$ gives, uniformly over $\beta \in [\beta_-, \beta_+]$,

$$R_T(\beta; C) \ge B 2^{-(T-1)} \binom{T-1}{\lfloor (T-1)/2 \rfloor} \ge \frac{B}{\sqrt{2T}} > 0. \tag{1}$$

Its supported no-carryover prior belongs to every $\mathcal{P}_{T,\beta}(C)$, and its likelihood argument covers every $\pi \in \mathcal{D}_T$, including deterministic, history-dependent, and outcome-adaptive kernels without positivity.

We first establish the improved finite selector. Draw one random variable $R \sim \mathrm{Bernoulli}(1/2)$ before outcomes, set $W_t = R$ for every t, and define

$$\widehat{\theta}^{\mathrm{hom}} = (2R-1)\frac{1}{T}\sum_{t=1}^{T} Y_t^{\mathrm{obs}}. \tag{2}$$

This outcome-independent rule is a member of $\mathcal{D}_T$. For a fixed array, put

$$\mu_a = \frac{1}{T}\sum_{t=1}^{T} Y_t(\mathbf{a}_{1:t}), \qquad a \in \{0,1\}.$$

By Definition $\mathrm{def:gate}$, $\mathrm{GATE}_T(Y) = \mu_1 - \mu_0$. If $R=1$, then $\widehat{\theta}^{\mathrm{hom}} = \mu_1$, so the absolute error is $|\mu_0|$. If $R=0$, then $\widehat{\theta}^{\mathrm{hom}} = -\mu_0$, so the absolute error is $|\mu_1|$. Consequently

$$\mathbb{E}\left|\widehat{\theta}^{\mathrm{hom}} - \mathrm{GATE}_T(Y)\right| = \frac{|\mu_0|+|\mu_1|}{2} \le B, \tag{3}$$

where the inequality follows from Assumption $\mathrm{ass:bounded-array}$. The constant array $Y_t(z_{1:t}) = B$ belongs to $\mathcal{P}_{T,\beta}(C)$: it is bounded and its historical differences vanish. For this array $\mu_0 = \mu_1 = B$, and equality holds in (3). Thus the exact worst-case risk of the persistent homogeneous-path fair-coin procedure is B, for every T, C, β. This calculation does not assert positivity for that branch.

At $T=1$, the same construction is $W_1 \sim \mathrm{Bernoulli}(1/2)$ with $\widehat{\theta}_1 = (2W_1-1)Y_1^{\mathrm{obs}}$, and (3) gives risk at most B. The lower bound in (1) equals B, proving

$$R_1(\beta; C) = B. \tag{4}$$

Suppose $T \ge 2$. For every $b \in \{1, \ldots, T\}$, Lemma $\mathrm{lem:markov-washout-oracle-upper}$ constructs the stated stationary Markov switchback and observed-data Horvitz–Thompson estimator and proves

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \mathbb{E}_{\pi^{(b)}} \left|\widehat{\vartheta}_b - \mathrm{GATE}_T(Y)\right| \le U^{\mathrm{cond}}_{T,\beta}(b). \tag{5}$$

Combining (3) and (5) yields

$$R_T(\beta;C) \le \min\left\{B,\ \min_{1\le b\le T} U^{\mathrm{cond}}_{T,\beta}(b)\right\} = \mathfrak{U}^{\mathrm{cond}}_{T,\beta}. \tag{6}$$

The index set is finite, so the Markov minimum is attained. One may select before assignment between the persistent homogeneous branch and the smallest minimizing Markov scale using only T, B, C, β. Together with (4), this proves the finite upper certificate for every T.

Put $K = 8e-4$, $x = T^{1/(2\beta+1)}$, and $b_\beta = \lceil x \rceil$. For $T \ge 2$, one has $1 < x < T$, hence $b_\beta \in \{1,\dots,T\}$, and

$$\frac{b_\beta}{x} \le 1 + \frac{1}{x} \le 1 + 2^{-1/(2\beta+1)}, \qquad b_\beta \ge x. \tag{7}$$

The final inequality of Lemma $\mathrm{lem:markov-washout-oracle-upper}$, together with $(T-b_\beta-1)_+/T \le 1$, $(b_\beta+1)^{-\beta} \le b_\beta^{-\beta}$, and (7), yields

$$\begin{aligned} U^{\mathrm{cond}}_{T,\beta}(b_\beta) &\le \sqrt{K}\,B\sqrt{\frac{b_\beta}{T}} + \frac{C}{\beta} b_\beta^{-\beta} \\ &\le \left[\sqrt{K\left(1+2^{-1/(2\beta+1)}\right)}\,B + \frac{C}{\beta}\right] T^{-\beta/(2\beta+1)}. \end{aligned} \tag{8}$$

At $T=1$, the piecewise certificate is B, and the coefficient in (8) is larger than B, so the same chain holds. Finally,

$$2^{-1/(2\beta+1)} \le 2^{-1/(2\beta_+ +1)}, \qquad \beta^{-1} \le \beta_-^{-1}. \tag{9}$$

Substituting $K = 8e-4$ into (8) and applying (9) gives the coefficient $C_0^\star$.

The Markov exposure floor, the exact worst-case auxiliary variance $B^2 V^\dagger_{T,b}$, the sharp uniform coefficient $8e-4$, and the exact conditional mismatch calculation all follow from Lemma $\mathrm{lem:markov-washout-oracle-upper}$. They apply only to the Markov branches. In particular, the conditional carryover term is an expected replacement-error bound rather than an exact bias or maximal error, and exactness of the auxiliary variance implies neither exact observed-data risk nor equality between the optimized certificate and oracle risk. Replacing the former $2B$ selector branch by (2) changes none of these facts and does not close the mismatch between the lower and upper exponents. □ □

Justification. The theorem combines the policy-uniform hidden-counterfactual-sign floor, the exact B-risk persistent homogeneous-path selector, the exact Markov auxiliary variance, and the conditional replacement certificate. *Gap.* The lower and upper exponents remain unmatched, so the exact growing-horizon oracle order is not identified. *Consumer.* With known β, compare the persistent homogeneous-path fair-coin procedure with the Markov certificates $U^{\mathrm{cond}}_{T,\beta}(b)$ and select the smallest certified branch before assignment.

Proposition 1 (prop:no-carryover-reduction)**.** *For every β, $\mathcal{P}_{T,\beta}(0) = \mathcal{P}^{(0)}_T$ literally: substituting $C=0$ in the historical-envelope definition forces equality exactly for path pairs with $z_t = z'_t$, while $Y_t(1, z_{1:t-1})$ and $Y_t(0, z'_{1:t-1})$ remain unrestricted apart from boundedness. Consequently $R_T(\beta;0) = R^{(0)}_T \asymp_B T^{-1/2}$. The upper construction is independent* Bernoulli(1/2) *assignment with the ordinary timewise positive-probability Horvitz–Thompson direct-effect contrast; a bounded fixed-array lower bound matches it.*

Proof. **Proof.** We first identify the two classes. Fix β and substitute $C=0$ in the definition of $\mathcal{P}_{T,\beta}(C)$. For any two paths with $z_t = z'_t$, the historical-envelope inequality becomes

$$|Y_t(z_{1:t}) - Y_t(z'_{1:t})| \le \sum_{\ell=1}^{t-1} 0 \cdot (1+\ell)^{-(1+\beta)} \mathbf{1}\{z_{t-\ell} \ne z'_{t-\ell}\} = 0.$$

Thus $Y_t(z_{1:t}) = Y_t(z'_{1:t})$ whenever the two paths have the same contemporaneous treatment. Together with the common bound $|Y_t(z_{1:t})| \le B$, this is precisely the definition of $\mathcal{P}^{(0)}_T$. Conversely, every $Y \in \mathcal{P}^{(0)}_T$ has a zero left-hand side for every path pair to which the historical-envelope restriction applies, and therefore belongs to $\mathcal{P}_{T,\beta}(0)$. Hence

$$\mathcal{P}_{T,\beta}(0) = \mathcal{P}^{(0)}_T.$$

The envelope compares only paths satisfying $z_t = z'_t$. It therefore imposes no equality between $Y_t(1, z_{1:t-1})$ and $Y_t(0, z'_{1:t-1})$, so direct effects remain unrestricted apart from boundedness. Since the definitions of the two minimax risks then have identical design classes, estimator classes, losses, targets, and adversarial classes, the class equality gives

$$R_T(\beta; 0) = R_T^{(0)}.$$

We next give the explicit upper bound. Let $W_1, \ldots, W_T$ be independent Bernoulli(1/2) assignments. This non-adaptive rule is an element of $\mathcal{D}_T$. For $Y \in \mathcal{P}_T^{(0)}$, let $y_{t,a}$ denote the common value of $Y_t(z_{1:t})$ among paths ending in $a \in \{0, 1\}$. Define

$$\widehat{\theta}_{\mathrm{HT}} = \frac{1}{T} \sum_{t=1}^{T} \left\{ \frac{W_t Y_t^{\mathrm{obs}}}{1/2} - \frac{(1 - W_t) Y_t^{\mathrm{obs}}}{1/2} \right\} = \frac{1}{T} \sum_{t=1}^{T} \{2W_t y_{t,1} - 2(1 - W_t) y_{t,0}\}.$$

The expectation of its t-th unscaled summand is $y_{t,1} - y_{t,0}$, so $\widehat{\theta}_{\mathrm{HT}}$ is unbiased for $\mathrm{GATE}_T(Y)$. Its centered t-th unscaled summand is

$$(2W_t - 1)(y_{t,1} + y_{t,0}).$$

Independence across times and $|y_{t,a}| \le B$ imply

$$\mathrm{Var}_\pi(\widehat{\theta}_{\mathrm{HT}}) = \frac{1}{T^2} \sum_{t=1}^{T} (y_{t,1} + y_{t,0})^2 \le \frac{4B^2}{T}.$$

Consequently, by Cauchy–Schwarz,

$$\sup_{Y \in \mathcal{P}_T^{(0)}} \mathbb{E}_\pi \left| \widehat{\theta}_{\mathrm{HT}} - \mathrm{GATE}_T(Y) \right| \le \frac{2B}{\sqrt{T}},$$

and hence $R_T^{(0)} \le 2BT^{-1/2}$.

For the lower bound, fix an arbitrary sequential kernel $\pi \in \mathcal{D}_T$ and an arbitrary observed-data estimator $\widehat{\theta}$. As a minimax proof device, draw mutually independent variables $X_{t,a}$, for $t = 1, \ldots, T$ and $a \in \{0, 1\}$, each uniformly from $\{-B, B\}$, before assignment, and set

$$Y_t(z_{1:t}) = X_{t, z_t}.$$

Every realized array is fixed during the experiment, is bounded by B, and is in $\mathcal{P}_T^{(0)}$. Thus this prior is supported on the finite-population adversarial class and does not add outcome noise conditional on the realized array.

Let $D = (W, Y^{\mathrm{obs}})$. Conditional on every observed-data value of positive probability, the unobserved coordinates $X_{t,1-W_t}$, $t = 1, \ldots, T$, remain mutually independent and uniform on $\{-B, B\}$. Indeed, the joint mass of the latent coordinates and the observed data factors into their independent prior masses, indicators fixing the observed coordinates, and

$$\prod_{t=1}^{T} \pi_t\big(W_t \mid W_{1:t-1}, Y_{1:t-1}^{\mathrm{obs}}\big).$$

The last factor is measurable with respect to the observed data and contains no unobserved coordinate. This argument permits randomized, deterministic, history-dependent, and outcome-adaptive kernels and uses no positivity condition.

Define the observed-data quantity

$$K(D)=\frac{1}{T}\sum_{t=1}^{T}(2W_t-1)Y_t^{\mathrm{obs}}.$$

It follows from the preceding conditional factorization that

$$\mathrm{GATE}_T(Y)\mid D \;\stackrel{d}{=}\; K(D)+\frac{B}{T}S_T, \qquad S_T=\sum_{t=1}^{T}\varepsilon_t,$$

where, conditionally on D, the variables $\varepsilon_1,\ldots,\varepsilon_T$ are independent Rademacher variables. For every real a, symmetry of S_T and the triangle inequality give

$$\mathbb{E}|a-S_T|=\frac{1}{2}\mathbb{E}\{|a-S_T|+|a+S_T|\}\geq \mathbb{E}|S_T|.$$

Applying this conditionally with $a=T\{\widehat{\theta}-K(D)\}/B$ yields

$$\mathbb{E}\left[\left|\widehat{\theta}-\mathrm{GATE}_T(Y)\right|\mid D\right]\geq \frac{B}{T}\mathbb{E}|S_T|.$$

We retain the exact Rademacher moment. If $J\sim\mathrm{Binomial}(T,1/2)$, then S_T has the same distribution as $2J-T$. With $r=\lfloor T/2\rfloor+1$, symmetry and

$$(2j-T)\binom{T}{j}=T\left\{\binom{T-1}{j-1}-\binom{T-1}{j}\right\}$$

give the telescoping identity

$$\mathbb{E}|S_T|=2^{1-T}\sum_{j=r}^{T}(2j-T)\binom{T}{j}$$
$$=T2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2\rfloor}.$$

For completeness, this exact expression is at least $\sqrt{T/2}$. To see this, put

$$p_k=4^{-k}\binom{2k}{k}, \qquad k\geq 1.$$

Since $p_1=1/2$ and

$$\frac{p_{k+1}}{p_k}=\frac{2k+1}{2k+2}\geq\sqrt{\frac{k}{k+1}},$$

induction gives $p_k\geq(2\sqrt{k})^{-1}$. If $n=2k$, then

$$2^{-n}\binom{n}{\lfloor n/2\rfloor}=p_k\geq\frac{1}{\sqrt{2n}}\geq\frac{1}{\sqrt{2(n+1)}};$$

if $n = 2k+1$, then

$$2^{-n}\binom{n}{\lfloor n/2 \rfloor} = p_{k+1} \geq \frac{1}{2\sqrt{k+1}} = \frac{1}{\sqrt{2(n+1)}}.$$

The same inequality is immediate for $n = 0$. Taking $n = T-1$ in the exact moment formula proves

$$\mathbb{E}|S_T| = T2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2 \rfloor} \geq \sqrt{\frac{T}{2}}.$$

After averaging over D, the Bayes risk under the supported prior is therefore at least $B/\sqrt{2T}$. The supremum risk is no smaller than this prior-average risk, so the arbitrary pair $(\pi, \widehat{\theta})$ satisfies

$$\sup_{Y \in \mathcal{P}_T^{(0)}} \mathbb{E}_\pi \left| \widehat{\theta} - \mathrm{GATE}_T(Y) \right| \geq \frac{B}{\sqrt{2T}}.$$

Taking the two infima gives $R_T^{(0)} \geq B/\sqrt{2T}$. Combining the bounds,

$$\frac{B}{\sqrt{2T}} \leq R_T(\beta; 0) = R_T^{(0)} \leq \frac{2B}{\sqrt{T}},$$

which proves $R_T(\beta; 0) = R_T^{(0)} \asymp_B T^{-1/2}$, with the stated explicit design and estimator. □ □

Justification. This proposition identifies the literal zero-radius member of the same class and risk definition without deleting direct effects. *Gap.* This is the required no-carryover sanity reduction, not a comparison against an external estimator. *Consumer.* It verifies that unrestricted immediate treatment effects retain ordinary root-T randomization precision when historical carryover is absent.

Proposition 2 (prop:finite-order-scope). *Fix $C > 0$. For every $m < T-1$, $\mathcal{P}_{T,\beta}(C) \cap \mathcal{F}_{T,m} \subsetneq \mathcal{P}_{T,\beta}(C)$. Let $a_m = \frac{1}{2}\min\{B, C(2+m)^{-(1+\beta)}\} > 0$ and set $Y_t(z_{1:t}) = a_m z_{t-m-1}$ for $t \geq m+2$ and $Y_t(z_{1:t}) = 0$ otherwise. This array is bounded, obeys the historical polynomial envelope, and is not in $\mathcal{F}_{T,m}$. Moreover $Y_t(z_{1:t}) = \sum_{\ell=1}^{t-1} \gamma_\ell z_{t-\ell}$, with $\gamma_\ell = C(1+\ell)^{-(1+\beta)}/\{2\max[1, (C/B)\sum_{u=1}^{T-1}(1+u)^{-(1+\beta)}]\}$, is bounded, admissible, has positive effects at every lag, and lies outside every prespecified $\mathcal{F}_{T,m}$.*

Proof. **Proof.** For the first witness,

$$a_m = \frac{1}{2}\min\{B, C(2+m)^{-(1+\beta)}\} > 0.$$

Thus $a_m \leq B/2$. If two histories with the same current treatment are compared at time t, the corresponding outcomes can differ only through the lag-$(m+1)$ coordinate, and therefore

$$|Y_t(z_{1:t}) - Y_t(z'_{1:t})| \leq a_m \mathbf{1}\{z_{t-m-1} \neq z'_{t-m-1}\} \leq C(2+m)^{-(1+\beta)}\mathbf{1}\{z_{t-m-1} \neq z'_{t-m-1}\}.$$

This is one term of the required historical envelope. At $t = m+2$, choose two histories that agree in coordinates $2, \ldots, m+2$ and differ in coordinate one. They agree on $z_{(t-m)\vee 1:t} = z_{2:m+2}$, but their outcomes differ by $a_m > 0$. Hence this array is in $\mathcal{P}_{T,\beta}(C)$ and not in $\mathcal{F}_{T,m}$.

For the second witness, put

$$S = \sum_{u=1}^{T-1}(1+u)^{-(1+\beta)}, \qquad D = 2\max\{1, (C/B)S\}.$$

Then $\gamma_\ell = C(1+\ell)^{-(1+\beta)}/D > 0$, and

$$0 \leq Y_t(z_{1:t}) \leq \sum_{\ell=1}^{T-1} \gamma_\ell = \frac{CS}{2\max\{1, (C/B)S\}} \leq \frac{B}{2}.$$

For equal-current histories, the triangle inequality gives

$$|Y_t(z_{1:t}) - Y_t(z'_{1:t})| \le \sum_{\ell=1}^{t-1} \gamma_\ell \mathbf{1}\{z_{t-\ell} \ne z'_{t-\ell}\} \le \sum_{\ell=1}^{t-1} C(1+\ell)^{-(1+\beta)} \mathbf{1}\{z_{t-\ell} \ne z'_{t-\ell}\},$$

because $D \ge 2$. Thus the array is bounded and admissible. For every prespecified $m < T-1$, the same two histories at $t = m+2$ produce outcomes differing by $\gamma_{m+1} > 0$, so this array is not in $\mathcal{F}_{T,m}$.

Finally, the zero array belongs to $\mathcal{P}_{T,\beta}(C) \cap \mathcal{F}_{T,m}$, whereas the first witness belongs to $\mathcal{P}_{T,\beta}(C)$ but not to $\mathcal{F}_{T,m}$. Therefore

$$\mathcal{P}_{T,\beta}(C) \cap \mathcal{F}_{T,m} \subsetneq \mathcal{P}_{T,\beta}(C).$$

No common-design or minimax-optimality consequence is used or asserted. □ □

Justification. The proposition makes the class relation with every prespecified finite-memory model literal using two explicit fixed arrays. *Gap.* It is a structural scope statement only and carries no common-design or minimax-optimality consequence. *Consumer.* A reader can verify whether a deployment lies in a finite-order intersection or requires the infinite-tail envelope.

Open-ended Question 1 (oeq:sharp-adaptation)**.** *For fixed $C > 0$ and $\beta_- < \beta_+$, determine the joint status of (K1), a fixed-before-assignment endpoint-prior construction proving a policy-uniform two-class lower bound $A_T(C) \ge g_T$ with $g_T \to \infty$, and (K2), one common multiscale design and estimator, independent of the realized β, proving $A_T(C) = T^{o(1)}$; then determine the sharp asymptotic order of $A_T(C)$. The two routes are not mutually exclusive; logarithmic divergence, for example, could satisfy both. Put*

$$C_0^\star = \sqrt{(8e-4)\left(1 + 2^{-1/(2\beta_+ + 1)}\right)}\, B + \frac{C}{\beta_-}.$$

For $b \in \{1, \dots, T\}$, define $r_b = b/(b+1)$, $k_t = \min\{b, t-1\}$, $\lambda_b = (b-1)/(b+1)$,

$$V^\dagger_{T,b} = 4 \sum_{t=1}^{T} r_b^{-k_t} + 8 \sum_{h=1}^{b} \sum_{t=1}^{T-h} r_b^{h - k_{t+h}} + 8 \sum_{d=1}^{(T-b-1)_+} (T-b-d)\lambda_b^d,$$

$$E^{\mathrm{cond}}_{T,\beta}(b) = \frac{2}{T} \sum_{t=b+2}^{T} \min\left\{2B, \frac{C}{2} \sum_{\ell=b+1}^{t-1} (1+\ell)^{-(1+\beta)}(1 - \lambda_b^{\ell-b})\right\}, \qquad U^{\mathrm{cond}}_{T,\beta}(b) = \frac{B}{T}\sqrt{V^\dagger_{T,b}} + E^{\mathrm{cond}}_{T,\beta}(b),$$

with empty inner sums equal to zero. Put $b_- = \lceil T^{1/(2\beta_- + 1)} \rceil$ and define

$$N_T^{\mathrm{fin}} = \begin{cases} B, & T = 1, \\ U^{\mathrm{cond}}_{T,\beta_-}(b_-), & T \ge 2, \end{cases} \qquad N_T^\star = \begin{cases} B, & T = 1, \\ \min\left\{B, \min_{1 \le b \le T} U^{\mathrm{cond}}_{T,\beta_-}(b)\right\}, & T \ge 2, \end{cases}$$

and

$$D_T^{\mathrm{bin}} = B 2^{-(T-1)} \binom{T-1}{\lfloor (T-1)/2 \rfloor}.$$

The strongest currently certified common-design polynomial upper certificate is

$$1 \le A_T(C) \le \frac{N_T^\star}{D_T^{\mathrm{bin}}} \le \frac{N_T^{\mathrm{fin}}}{D_T^{\mathrm{bin}}} \le \frac{\sqrt{2}}{B}\left[\sqrt{(8e-4)\left(1 + 2^{-1/(2\beta_- + 1)}\right)}\, B + \frac{C}{\beta_-}\right] T^{1/(4\beta_- + 2)}.$$

At $T = 1$, $N_1^\star = N_1^{\mathrm{fin}} = D_1^{\mathrm{bin}} = B$ and $A_1(C) = 1$, using the explicit fair-Bernoulli one-period procedure rather than an assertion that $U^{\mathrm{cond}}_{1,\beta}(1) = B$. For $T \ge 2$, the B branch draws $R \sim \mathrm{Bernoulli}(1/2)$ before outcomes, sets $W_t = R$, and uses $\widehat{\theta}^{\mathrm{hom}} = (2R-1)T^{-1}\sum_{t=1}^{T} Y_t^{\mathrm{obs}}$. Its fixed-array risk is $(|\mu_0| + |\mu_1|)/2 \le B$, where $\mu_a = T^{-1}\sum_t Y_t(\mathbf{a}_{1:t})$, and the admissible constant array attains B. The common selector chooses before assignment between this persistent *homogeneous-path fair-coin branch and the smallest minimizing β_--tuned Markov scale, independently of the realized β. The ratios use the exact binomial Bayes lower bound as the oracle-risk denominator; they do not use the unknown exact oracle risk except at $T = 1$, and the certificate is not K2. The endpoint-specific coefficient is retained because β_- is known for this common design; it is not replaced by the larger β_+-uniform constant $C_0^\star$. The certified two-class factor remains $g_T = 1$. The proved supported-prior lower obstructions are policy-uniform within their individual classes, but no policy-uniform two-class cross-law tradeoff has been derived. Neither a divergent g_T, a $T^{o(1)}$ common multiscale procedure, a matching oracle rate, nor the sharp order has been proved; K1 and K2 remain logically compatible.*

The strongest closed finite-horizon conclusion, realigned to the current exact-B selector, is

$$1 \le A_T(C) \le \frac{N_T^\star}{D_T^{\mathrm{bin}}} \le \frac{N_T^{\mathrm{fin}}}{D_T^{\mathrm{bin}}} \le \frac{\sqrt{2}}{B}\left[\sqrt{(8e-4)\left(1+2^{-1/(2\beta_-+1)}\right)}\,B + \frac{C}{\beta_-}\right] T^{1/(4\beta_-+2)}.$$

For $T \ge 2$,

$$N_T^\star = \min\left\{B, \min_{1\le b\le T} U_{T,\beta_-}^{\mathrm{cond}}(b)\right\},$$

and the B branch is the fixed-before-assignment persistent homogeneous-path fair-coin procedure. For every fixed array its risk is $(|\mu_0| + |\mu_1|)/2 \le B$, and the admissible constant array attains B, so that branch has exact worst-case risk B. Moreover, $A_1(C) = 1$, while the strongest unconditional two-class factor remains $g_T = 1$. The proved low-energy and endpoint-persistence lower bounds remain valid, but they do not imply K1 or K2; bounded adaptation, logarithmic divergence satisfying both K1 and K2, and polynomial regimes remain compatible with the declared results.

Justification. The B-elbow improves the finite common-design numerator but does not couple the rough and smooth supported-prior obstructions across decay classes. *Gap.* A divergent two-class factor, a common $T^{o(1)}$ multiscale procedure, a matching oracle rate, and the sharp order of $A_T(C)$ remain open. *Consumer.* A resolution would quantify the growing-horizon value of knowing the decay index beyond the exact finite selector and endpoint-specific polynomial certificate.

Lemma 1 (lem:rough-prior-cadence-kl)**.** *Fix $C > 0$, $\beta \in [\beta_-, \beta_+]$, and an integer b with $2b \le T$. Put $a = B/2$, choose $\eta \in (0, 1/2]$ such that $\eta \sum_{\ell=b}^{2b-1} C(1+\ell)^{-(1+\beta)} \le a/4$, and let $U_1, \dots, U_T$ be independent* $\mathrm{Unif}[-a, a]$ *variables drawn before assignment. Define $\psi_q(u) = u + q(1 - u^2/a^2)$. Let Π_0 put mass on arrays $Y_t(z_{1:t}) = U_t$. Let Π_1 put mass on arrays*

$$Y_t(z_{1:t}) = \psi_{q_t(z_{1:t})}(U_t), \qquad q_t(z_{1:t}) = \eta \sum_{\ell=b}^{2b-1} C(1+\ell)^{-(1+\beta)}(2z_{t-\ell} - 1)$$

for $t \ge 2b$, with $q_t = 0$ for $t < 2b$. Both priors are fixed before assignment and are supported on $\mathcal{P}_{T,\beta}(C)$. If $P_{j,\pi}$ is the observed-history law induced by Π_j and any $\pi \in \mathcal{D}_T$, then

$$\mathrm{KL}(P_{1,\pi} \| P_{0,\pi}) \le \frac{4}{3a^2}\mathbb{E}_{P_{1,\pi}} \sum_{t=2b}^{T} q_t(W_{1:t})^2.$$

Moreover, writing $s_b = \eta \sum_{\ell=b}^{2b-1} C(1+\ell)^{-(1+\beta)}$, the causal target equals zero under Π_0, whereas under Π_1,

$$\mathrm{GATE}_T(Y) = \frac{2s_b}{T} \sum_{t=2b}^{T} (1 - U_t^2/a^2).$$

Proof. **Proof.** First, $\psi_q(-a) = -a$, $\psi_q(a) = a$, and

$$\frac{\partial}{\partial u}\psi_q(u) = 1 - \frac{2qu}{a^2} \in [1/2, 3/2]$$

whenever $|q| \le a/4$ and $|u| \le a$. Thus ψ_q maps $[-a, a]$ increasingly onto itself, so every realized array is bounded by $a \le B$. Under Π_1, $|q_t| \le a/4$. If two histories with the same current assignment differ at lag $\ell \in \{b, \dots, 2b-1\}$, the corresponding values of q_t differ by at most $2\eta C(1+\ell)^{-(1+\beta)}$. Since $|\psi_q(u)-\psi_{q'}(u)|\le|q-q'|$, changing any collection of historical coordinates changes the outcome by at most the sum of their envelope weights because $2\eta\le 1$. Coordinates outside this lag band have no effect. Hence every realization under Π_1 belongs to $\mathcal{P}_{T,\beta}(C)$; membership under Π_0 is immediate.

Conditional on the pre-outcome history at time t, the quantity $q=q_t(W_{1:t})$ is fixed and U_t is still uniform on $[-a,a]$. If $V=\psi_q(U_t)$, change of variables and the displayed derivative give

$$\mathrm{KL}(\mathcal{L}(V)\|\,\mathrm{Unif}[-a,a])=\mathbb{E}\left[-\log\left(1-\frac{2qU_t}{a^2}\right)\right].$$

For $|x|\le 1/2$, direct differentiation shows $-\log(1-x)\le x+x^2$. Since $\mathbb{E}U_t=0$ and $\mathbb{E}U_t^2=a^2/3$, the last display is at most $4q^2/(3a^2)$. In the likelihood ratio for the complete observed history, the sequential assignment kernel is the same measurable function of past observations under the two priors and therefore cancels. The chain rule for densities then sums the conditional bounds, proving the policy-uniform Kullback–Leibler inequality.

Finally, the all-treated history has $q_t=s_b$, while the all-control history has $q_t=-s_b$, for every $t\ge 2b$. Therefore

$$\psi_{s_b}(U_t)-\psi_{-s_b}(U_t)=2s_b(1-U_t^2/a^2).$$

The early terms vanish, and averaging this identity proves the target formula. Under Π_0, every contemporaneous contrast is zero. □ □

Lemma 2 (lem:smooth-block-prior-risk)**.** *Fix an integer $L\in\{1,\dots,T\}$, put $n=\lfloor T/L\rfloor$, and partition the first nL times into consecutive blocks $I_1,\dots,I_n$ of length L. Draw independent Rademacher variables $\xi_{k,0},\xi_{k,1}$ before assignment and define $Y_t(z_{1:t})=B\xi_{k,z_t}$ for $t\in I_k$, with $Y_t=0$ for $t>nL$. This prior is supported on the no-carryover subclass of $\mathcal{P}_{T,\beta_+}(C)$. For every $\pi\in\mathcal{D}_T$ and every observed-data estimator $\widehat{\theta}$, if $U_L(W)$ denotes the number of blocks I_k on which only one arm occurs, then*

$$\mathbb{E}_{\pi,\xi}\left|\widehat{\theta}-\mathrm{GATE}_T(Y)\right|\ge\frac{BL}{\sqrt{3}\,T}\mathbb{E}_{\pi,\xi}\sqrt{U_L(W)}.$$

Proof. **Proof.** Every realized array is bounded by B and depends on a treatment history only through its current assignment. It therefore belongs to the no-carryover subclass of every polynomial-envelope class, in particular $\mathcal{P}_{T,\beta_+}(C)$.

Fix the complete observed history. If arm a never occurs in block I_k, then $\xi_{k,a}$ appears in no observed outcome and in no assignment-kernel factor: before an arm is selected, the sequential policy cannot depend on its unobserved sign. Consequently, conditional on the complete data, every such missing-arm sign remains an independent Rademacher variable. On a monochromatic block exactly one of the two signs is missing. The target can thus be written conditionally as

$$\mathrm{GATE}_T(Y)=M(W,Y^{\mathrm{obs}})+\frac{BL}{T}\sum_{j=1}^{U_L(W)}\varepsilon_j,$$

where M is observed-data measurable and the ε_j are conditionally independent Rademacher variables; signs from any additional unobserved components can only add independent symmetric terms and may be retained in the same sum. For absolute loss, the conditional posterior median of the centered symmetric sum is zero. If $S_u=\sum_{j=1}^{u}\varepsilon_j$, then $\mathbb{E}S_u^2=u$ and $\mathbb{E}S_u^4=3u^2-2u\le 3u^2$. Hölder's inequality gives $\mathbb{E}|S_u|\ge\sqrt{u/3}$. Hence every estimator has conditional Bayes risk at least $BL\sqrt{U_L(W)}/(\sqrt{3}T)$. Averaging over the observed history proves the result. □ □

Lemma 3 (lem:exact-positive-oracle-risk)**.** *For every $C\ge 0$, $\beta\in[\beta_-,\beta_+]$, and $T\ge 1$,*

$$R_T(\beta;C)\ge B2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2\rfloor}\ge\frac{B}{\sqrt{2T}}.$$

The first lower bound is the exact Bayes absolute risk contributed by the independent hidden counterfactual signs in the supported Rademacher prior.

Proof. **Proof.** Fix an arbitrary $\pi \in \mathcal{D}_T$ and an arbitrary observed-data estimator $\widehat{\theta}$. Draw independent Rademacher signs $(\xi_{t,0}, \xi_{t,1})_{t=1}^T$ before assignment and define the fixed potential-outcome array

$$Y_t^\xi(z_{1:t}) = B\xi_{t,z_t}.$$

Every realization is bounded by B and depends only on the contemporaneous treatment. Hence, whenever two paths have the same final coordinate, their outcomes agree; the historical-envelope inequality holds with left side zero. Thus every realization belongs to $\mathcal{P}_{T,\beta}(C)$ for every $C \geq 0$ and every admissible β.

Under this prior followed by the sequential design, the likelihood contribution at time t is

$$\pi_t(w_t \mid w_{1:t-1}, y_{1:t-1}^{\text{obs}})\mathbf{1}\{y_t^{\text{obs}} = B\xi_{t,w_t}\}.$$

No factor contains the unobserved sign $\xi_{t,1-w_t}$. Therefore, conditional on the complete observed data (W, Y^{obs}), the variables

$$\varepsilon_t = (1 - 2W_t)\xi_{t,1-W_t}, \qquad t = 1, \dots, T,$$

remain independent Rademacher signs. This likelihood cancellation is valid even when later assignment probabilities depend on earlier observed outcomes.

The causal estimand decomposes as

$$\text{GATE}_T(Y^\xi) = M(W, Y^{\text{obs}}) + \frac{B}{T}\sum_{t=1}^T \varepsilon_t,$$

where

$$M(W, Y^{\text{obs}}) = \frac{B}{T}\sum_{t=1}^T (2W_t - 1)\xi_{t,W_t}$$

is observed-data measurable. If $S_T = \sum_{t=1}^T \varepsilon_t$, its conditional distribution is symmetric about zero. Hence $M(W, Y^{\text{obs}})$ is a conditional median of the target, and the median characterization of absolute-loss minimizers gives

$$\mathbb{E}\left[\left|\widehat{\theta} - \text{GATE}_T(Y^\xi)\right| \,\middle|\, W, Y^{\text{obs}}\right] \geq \frac{B}{T}\mathbb{E}|S_T|.$$

The expectation on the right does not depend on the realized data.

We now evaluate it exactly. Let $N_T \sim \text{Binomial}(T, 1/2)$, so $S_T = 2N_T - T$ in distribution. Pairing the binomial masses on the two sides of zero and using $j\binom{T}{j} = T\binom{T-1}{j-1}$ yields

$$\mathbb{E}|S_T| = T2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2 \rfloor}.$$

For completeness, the constant bound follows elementarily. Put

$$p_k = 4^{-k}\binom{2k}{k} = \prod_{j=1}^k \frac{2j-1}{2j}.$$

The claim $p_k \geq (2\sqrt{k})^{-1}$ holds at $k = 1$. If it holds at k, then

$$p_{k+1} = p_k\frac{2k+1}{2k+2} \geq \frac{1}{2\sqrt{k}}\frac{2k+1}{2k+2} \geq \frac{1}{2\sqrt{k+1}},$$

because $(2k+1)^2 \geq 4k(k+1)$. Thus the claim holds for all $k \geq 1$. If $T = 2k$, the exact identity gives

$$\mathbb{E}|S_T| = 2kp_k \geq \sqrt{k} = \sqrt{T/2}.$$

If $T = 2k + 1$ with $k \geq 1$, it gives

$$\mathbb{E}|S_T| = (2k+1)p_k \geq \frac{2k+1}{2\sqrt{k}} \geq \sqrt{\frac{2k+1}{2}} = \sqrt{T/2}.$$

For $T = 1$, the same inequality follows directly from $\mathbb{E}|S_1| = 1$.

Averaging the conditional risk bound over the data and prior, and then using that the supremum over fixed arrays dominates the prior average, gives

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \mathbb{E}_\pi \left|\widehat{\theta} - \mathrm{GATE}_T(Y)\right| \geq \frac{B}{T}\mathbb{E}|S_T| = B2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2 \rfloor} \geq \frac{B}{\sqrt{2T}}.$$

Taking the infima over $\widehat{\theta}$ and π proves the result. □ □

Justification. This lemma exposes the exact supported-prior Bayes obstruction needed to sharpen both the oracle-risk floor and the adaptation denominator. *Gap.* It is a universal root-T lower bound and does not by itself exploit carryover decay to match the oracle upper rate. *Consumer.* The oracle theorem, positivity lemma, and common-design adaptation bracket use this exact lower bound.

Lemma 4 (lem:endpoint-certificate-gap)**.** *Fix $C > 0$ and $\beta \in [\beta_-, \beta_+]$. Let d be an integer satisfying $d \geq 4\,2^{1+\beta_+}$, let b be a positive multiple of d, and suppose $T \geq 4b$. Define the deterministic path $w \in \{0,1\}^T$ by $w_j = 0$ exactly when d divides j. Let $U_b(w)$ be the number of aligned blocks of length b among the first $\lfloor T/b \rfloor b$ coordinates on which only one arm occurs. For $\eta > 0$, put*

$$q_t(w) = \eta \sum_{\ell=b}^{2b-1} C(1+\ell)^{-(1+\beta)}(2w_{t-\ell} - 1), \qquad s_b = \eta \sum_{\ell=b}^{2b-1} C(1+\ell)^{-(1+\beta)}.$$

Then

$$U_b(w) = 0, \qquad q_t(w) \geq \frac{s_b}{2} \quad (2b \leq t \leq T), \qquad \sum_{t=2b}^{T} q_t(w)^2 \geq \frac{T}{8}s_b^2.$$

Thus the path functionals appearing in Lemmas $\mathrm{lem:rough-prior-cadence-kl}$ *and* $\mathrm{lem:smooth-block-prior-risk}$ *are not complementary: the smooth-prior certificate can vanish while the rough-prior likelihood-energy upper bound is of order Ts_b^2.*

Proof. **Proof.** Every aligned block of length b contains exactly b/d indices divisible by d and $b - b/d$ indices not divisible by d. Since $d > 1$, both arms occur in every such block, and therefore $U_b(w) = 0$.

Write $a_\ell = (1+\ell)^{-(1+\beta)}$ and $A = \sum_{\ell=b}^{2b-1} a_\ell$. Partition $\{b, \ldots, 2b-1\}$ into b/d consecutive groups of d indices. For any such group with first index $r \geq b$, monotonicity of a_ℓ, the inequality $d \leq b$, and $\beta \leq \beta_+$ give

$$\frac{\max_{r \leq \ell < r+d} a_\ell}{\min_{r \leq \ell < r+d} a_\ell} = \left(\frac{r+d}{r+1}\right)^{1+\beta} < 2^{1+\beta} \leq 2^{1+\beta_+}.$$

For each fixed t, exactly one index ℓ in every group satisfies $d \mid (t-\ell)$, and this is exactly the index for which $w_{t-\ell} = 0$. If N_t denotes the sum of the weights a_ℓ over those exceptional indices, the preceding ratio bound implies, group by group,

$$N_t \leq \frac{2^{1+\beta_+}}{d}A \leq \frac{A}{4}.$$

Consequently

$$q_t(w) = \eta C\,(A - 2N_t) \geq \frac{\eta C A}{2} = \frac{s_b}{2}$$

for every $t \in \{2b, \ldots, T\}$. Finally, $T \geq 4b$ implies $T - 2b + 1 \geq T/2$, whence

$$\sum_{t=2b}^{T} q_t(w)^2 \geq (T - 2b + 1)\frac{s_b^2}{4} \geq \frac{T}{8}s_b^2.$$

This proves every assertion. □ □

Lemma 5 (lem:positive-oracle-risk)**.** *For every $C \geq 0$, $\beta \in [\beta_-, \beta_+]$, and $T \geq 1$,*

$$R_T(\beta; C) \geq B2^{-(T-1)} \binom{T-1}{\lfloor (T-1)/2 \rfloor} \geq \frac{B}{\sqrt{2T}} > 0.$$

Consequently $A_T(C) \geq 1$.

Proof. **Proof.** Lemma lem : exact − positive − oracle − risk gives

$$R_T(\beta; C) \geq B2^{-(T-1)} \binom{T-1}{\lfloor (T-1)/2 \rfloor} \geq \frac{B}{\sqrt{2T}} > 0.$$

For any fixed $(\pi, \widehat{\theta})$, its worst-case risk over $\mathcal{P}_{T,\beta}(C)$ is at least the infimum $R_T(\beta; C)$. Division is valid by the displayed strict positivity, so the corresponding risk ratio is at least one for every β. Taking the supremum over β and then the infimum over $(\pi, \widehat{\theta})$ gives $A_T(C) \geq 1$. □ □

Justification. The supported Rademacher prior leaves one counterfactual sign unobserved at every time under every sequential design and now retains its exact absolute moment. *Gap.* There is no gap in positivity: the proof yields the stronger bound $R_T(\beta; C) \geq B/\sqrt{2T}$. This root-$T$ obstruction does not match the beta-aware upper exponent. *Consumer.* It validates every risk ratio in $A_T(C)$ and supplies the universal denominator lower bound used in the common-design comparison.

Lemma 6 (lem:rough-prior-low-energy-risk)**.** *Fix $C > 0$, $\beta \in [\beta_-, \beta_+]$, an integer b with $4b \leq T$, and $\pi \in \mathcal{D}_T$. In the construction of Lemma* lem : rough − prior − cadence − kl*, take $a = B/2$, let $\eta \in (0, 1/2]$ satisfy its stated amplitude condition, and let $\Pi_0, \Pi_1, P_{j,\pi}, q_t$, and s_b denote the resulting objects. If*

$$\mathbb{E}_{P_{1,\pi}} \sum_{t=2b}^{T} q_t(W_{1:t})^2 \leq \frac{3a^2}{64},$$

then every observed-data estimator $\widehat{\theta}$ satisfies

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \mathbb{E}_\pi \left| \widehat{\theta} - \mathrm{GATE}_T(Y) \right| \geq \frac{s_b}{25}.$$

Proof. **Proof.** Put $N = T - 2b + 1$ and write $\Theta = \mathrm{GATE}_T(Y)$. Under Π_0, Lemma lem : rough − prior − cadence − kl gives $\Theta = 0$. Under Π_1, it gives

$$\Theta = \frac{2s_b}{T} \sum_{t=2b}^{T} X_t, \qquad X_t = 1 - \frac{U_t^2}{a^2}.$$

The variables X_t are independent, and direct integration for $U_t/a \sim \mathrm{Unif}[-1, 1]$ yields

$$\mathbb{E}X_t = \frac{2}{3}, \qquad \mathrm{Var}(X_t) = \frac{4}{45}.$$

Consequently, if $\mu = \mathbb{E}_{\Pi_1}\Theta$, then

$$\mu = \frac{4s_b N}{3T}, \qquad \mathrm{Var}_{\Pi_1}(\Theta) = \frac{16 s_b^2 N}{45 T^2}.$$

Because $T \geq 4b$, one has $N \geq T/2$, so $\mu \geq 2s_b/3$; moreover $N \geq 3$. Chebyshev's inequality therefore gives

$$\Pr_{\Pi_1}\left\{\Theta < \frac{\mu}{2}\right\} \leq \frac{4\,\mathrm{Var}_{\Pi_1}(\Theta)}{\mu^2} = \frac{4}{5N}.$$

Let $P_0 = P_{0,\pi}$ and $P_1 = P_{1,\pi}$ be the two observed-history laws. The Kullback–Leibler bound in Lemma lem : rough − prior − cadence − kl and the assumed energy inequality imply

$$\mathrm{KL}(P_1 \| P_0) \leq \frac{4}{3a^2} \cdot \frac{3a^2}{64} = \frac{1}{16}.$$

For completeness, this entails $\|P_1 - P_0\|_{\mathrm{TV}} \le 1/4$ without invoking any additional result. Indeed, take densities p_1, p_0 with respect to $P_1 + P_0$. The scalar inequality $-\log x \ge 2(1-\sqrt{x})$ gives

$$\int (\sqrt{p_1} - \sqrt{p_0})^2 \le \mathrm{KL}(P_1 \| P_0).$$

Cauchy–Schwarz and $\int (\sqrt{p_1} + \sqrt{p_0})^2 \le 4$ then give

$$\|P_1 - P_0\|_{\mathrm{TV}} = \frac{1}{2} \int |p_1 - p_0| \le \sqrt{\mathrm{KL}(P_1 \| P_0)} \le \frac{1}{4}.$$

Let

$$r_j = \mathbb{E}_{\Pi_j, \pi} \left| \widehat{\theta} - \Theta \right|, \qquad j \in \{0,1\},$$

and define the observed-data test $\varphi = \mathbf{1}\{\widehat{\theta} \ge \mu/4\}$. Since $\Theta = 0$ under Π_0, Markov's inequality gives

$$P_0(\varphi = 1) \le \frac{4r_0}{\mu}.$$

Under Π_1, on the event $\{\Theta \ge \mu/2, \varphi = 0\}$ one has $|\widehat{\theta} - \Theta| \ge \mu/4$. Hence

$$P_1(\varphi = 0) \le \frac{4}{5N} + \frac{4r_1}{\mu}.$$

For every test event, the sum of its two errors is at least one minus total variation. Combining the preceding displays yields

$$\frac{4(r_0 + r_1)}{\mu} + \frac{4}{5N} \ge 1 - \|P_1 - P_0\|_{\mathrm{TV}} \ge \frac{3}{4}.$$

It follows that

$$\max\{r_0, r_1\} \ge \frac{\mu}{8} \left(\frac{3}{4} - \frac{4}{5N} \right) \ge \frac{2s_b/3}{8} \cdot \frac{29}{60} = \frac{29 s_b}{720} > \frac{s_b}{25}.$$

Both priors are supported on $\mathcal{P}_{T,\beta}(C)$ by Lemma lem : rough − prior − cadence − kl. The worst-case risk over fixed arrays dominates each of the two Bayes risks, and hence dominates their maximum. This proves the claim. □ □

Lemma 7 (lem:endpoint-bump-persistence-risk)**.** *Fix $C > 0$, $\beta \in [\beta_-, \beta_+]$, an integer K with $1 \le K < T$, a sequential design $\pi \in \mathcal{D}_T$, an estimator $\widehat{\theta}$, and an arm $\alpha \in \{0,1\}$. Let Q_π be the assignment-path law generated by π when $Y_t^{\mathrm{obs}} = 0$ at every time, and let*

$$E_{\alpha,K} = \{\textit{there is a } t \in \{K+1, \ldots, T\} \textit{ such that } W_{t-K:t} = \boldsymbol{\alpha}\}.$$

Set

$$\delta_K = \frac{1}{2} \min\left\{ B, C(1+K)^{-(1+\beta)} \right\}.$$

Then

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \mathbb{E}_\pi \left| \widehat{\theta} - \mathrm{GATE}_T(Y) \right| \ge \frac{(T-K)\delta_K}{2T} \left\{ 1 - Q_\pi(E_{\alpha,K}) \right\}.$$

Proof. **Proof.** Let Y^0 be the identically zero array. For $t > K$, define

$$Y_t^{\alpha,K}(z_{1:t}) = \mathbf{1}\{z_t = \alpha\} \left[\delta_K - \sum_{\ell=1}^{K} C(1+\ell)^{-(1+\beta)} \mathbf{1}\{z_{t-\ell} \ne \alpha\} \right]_+,$$

and put $Y_t^{\alpha,K} = 0$ for $t \le K$. Both arrays are fixed before assignment. The bump array takes values in $[0, \delta_K] \subseteq [-B, B]$. If two paths have the same current assignment different from α, both displayed outcomes are zero. If their common current assignment is α, the positive-part map is one-Lipschitz, and therefore

$$\left|Y_t^{\alpha,K}(z_{1:t}) - Y_t^{\alpha,K}(z'_{1:t})\right| \le \sum_{\ell=1}^{K} C(1+\ell)^{-(1+\beta)} \mathbf{1}\{z_{t-\ell} \ne z'_{t-\ell}\}.$$

Thus $Y^0, Y^{\alpha,K} \in \mathcal{P}_{T,\beta}(C)$.

Along the all-α path the bump equals δ_K at each $t > K$, whereas it is zero when the current treatment is $1-\alpha$. Hence

$$\mathrm{GATE}_T(Y^0) = 0, \qquad \left|\mathrm{GATE}_T(Y^{\alpha,K})\right| = \Delta_K, \qquad \Delta_K = \frac{T-K}{T}\delta_K.$$

Also, for $1 \le \ell \le K$,

$$C(1+\ell)^{-(1+\beta)} \ge C(1+K)^{-(1+\beta)} \ge 2\delta_K.$$

It follows that an observed outcome under $Y^{\alpha,K}$ can be nonzero only when $W_{t-K:t} = \boldsymbol{\alpha}$; on that event it equals δ_K.

Couple the experiments under Y^0 and $Y^{\alpha,K}$ by using the same auxiliary randomizer to sample each binary assignment from the common kernel π_t whenever their observed histories agree. Until $E_{\alpha,K}$ occurs, both observed-outcome histories are identically zero, so the coupled assignment and outcome histories agree. Under their common history before separation, the assignment path has law Q_π. Consequently the two complete observed data sets agree with probability at least $1 - Q_\pi(E_{\alpha,K})$.

On the event of agreement, the estimator takes one common value, say h, in the two experiments. The triangle inequality gives

$$|h - \mathrm{GATE}_T(Y^0)| + |h - \mathrm{GATE}_T(Y^{\alpha,K})| \ge \Delta_K.$$

Taking expectations in the coupling shows that the sum of the two design risks is at least $\Delta_K\{1 - Q_\pi(E_{\alpha,K})\}$. Their maximum is at least half this quantity, and the supremum over $\mathcal{P}_{T,\beta}(C)$ dominates that maximum. Substituting the value of Δ_K proves the result. □ □

Lemma 8 (lem:markov-washout-oracle-upper). *Fix $C > 0$, $\beta \in [\beta_-, \beta_+]$, and $1 \le b \le T$. Let $\pi^{(b)} \in \mathcal{D}_T$ initialize $W_1 \sim \mathrm{Bernoulli}(1/2)$ and, for $t \ge 2$, switch arms with probability $1/(b+1)$, independently of outcomes conditional on W_{t-1}. Put $r_b = b/(b+1)$, $k_t = \min\{b, t-1\}$, $H_{t,b,a} = \mathbf{1}\{W_{t-k_t:t} = \mathbf{a}\}$, and $\upsilon_{t,b,a} = \Pr_{\pi^{(b)}}(H_{t,b,a} = 1) = \frac{1}{2} r_b^{k_t}$. Define $q_b = (1+1/b)^b$ and $\lambda_b = (b-1)/(b+1)$. Then*

$$\upsilon_{t,b,a} \ge \frac{1}{2q_b} > \frac{1}{2e}.$$

The infimum over all admissible T, b, t, a is $1/(2e)$, approached but not attained as $b \to \infty$ along indices with $k_t = b$. Put

$$V_{T,b}^{\dagger} = 4\sum_{t=1}^{T} r_b^{-k_t} + 8\sum_{h=1}^{b}\sum_{t=1}^{T-h} r_b^{h-k_{t+h}} + 8\sum_{d=1}^{(T-b-1)_+} (T-b-d)\lambda_b^d,$$

with empty sums equal to zero. The explicit observed-data estimator

$$\widehat{\vartheta}_b = \frac{1}{T}\sum_{t=1}^{T}\left\{\frac{H_{t,b,1}Y_t^{\mathrm{obs}}}{\upsilon_{t,b,1}} - \frac{H_{t,b,0}Y_t^{\mathrm{obs}}}{\upsilon_{t,b,0}}\right\}$$

has positive denominators. With $y_{t,a} = Y_t(\mathbf{a}_{1:t})$,

$$Z_t = \frac{H_{t,b,1} y_{t,1}}{v_{t,b,1}} - \frac{H_{t,b,0} y_{t,0}}{v_{t,b,0}}, \qquad \widetilde{\theta}_b = \frac{1}{T} \sum_{t=1}^{T} Z_t,$$

one has

$$\mathbb{E} Z_t^2 \le 4B^2 r_b^{-k_t}.$$

If $u = t + h$ with $1 \le h \le b$ and $\rho = 2r_b^{h-k_u}$, then $\rho \ge 2$ and

$$\operatorname{Cov}(Z_t, Z_u) = (\rho - 1)(y_{t,1} y_{u,1} + y_{t,0} y_{u,0}) + y_{t,1} y_{u,0} + y_{t,0} y_{u,1},$$

so

$$|\operatorname{Cov}(Z_t, Z_u)| \le 4B^2 r_b^{h-k_u}.$$

The diagonal and near-pair bounds, and the far-pair bound $|\operatorname{Cov}(Z_t, Z_u)| \le 4B^2 \lambda_b^{u-b-t}$ for $u - t > b$, are attained simultaneously by $y_{t,0} = y_{t,1} = B$. Consequently

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \operatorname{Var}_{\pi^{(b)}} \left(\sum_{t=1}^{T} Z_t \right) = B^2 V_{T,b}^{\dagger}, \qquad \mathbb{E} \left| \widetilde{\theta}_b - \mathrm{GATE}_T(Y) \right| \le \frac{B}{T} \sqrt{V_{T,b}^{\dagger}}.$$

Thus $B^2 V_{T,b}^{\dagger}$ is the exact worst-case auxiliary homogeneous-history variance, not an exact estimator risk. Moreover,

$$V_{T,b}^{\dagger} \le 4q_b T + 8Tb(q_b - 1) + 4(T - b)(b - 1) < (8e - 4)Tb,$$

$$\lim_{b \to \infty} \lim_{T \to \infty} \frac{V_{T,b}^{\dagger}}{Tb} = 8e - 4,$$

and hence $8e - 4$ is the smallest coefficient in a uniform inequality $V_{T,b}^{\dagger} \le KTb$. For $t \ge b + 2$, $b + 1 \le \ell \le t - 1$, and $a \in \{0, 1\}$, time reversibility and conditional independence at $W_{t-b} = a$ give

$$\Pr_{\pi^{(b)}}(W_{t-\ell} \ne a \mid H_{t,b,a} = 1) = \frac{1 - \lambda_b^{\ell - b}}{2}.$$

Define

$$E_{T,\beta}^{\mathrm{cond}}(b) = \frac{2}{T} \sum_{t=b+2}^{T} \min \left\{ 2B, \frac{C}{2} \sum_{\ell=b+1}^{t-1} (1 + \ell)^{-(1+\beta)} (1 - \lambda_b^{\ell - b}) \right\}$$

and

$$U_{T,\beta}^{\mathrm{cond}}(b) = \frac{B}{T} \sqrt{V_{T,b}^{\dagger}} + E_{T,\beta}^{\mathrm{cond}}(b).$$

Then

$$\mathbb{E} |\widehat{\vartheta}_b - \widetilde{\theta}_b| \le E_{T,\beta}^{\mathrm{cond}}(b) \le \frac{C}{T} \sum_{\ell=b+1}^{T-1} (T - \ell)(1 + \ell)^{-(1+\beta)} (1 - \lambda_b^{\ell - b}) \le \frac{C}{\beta} \frac{(T - b - 1)_+}{T} (b + 1)^{-\beta}.$$

Replacement is zero through $t = b + 1$; empty sums make the formulas valid at $T = 1$, $b = 1$, and $b = T$. Finally,

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \mathbb{E}_{\pi^{(b)}} \left| \widehat{\vartheta}_b - \mathrm{GATE}_T(Y) \right| \le U_{T,\beta}^{\mathrm{cond}}(b) \le \sqrt{8e - 4}\, B \sqrt{\frac{b}{T}} + \frac{C}{\beta} \frac{(T - b - 1)_+}{T} (b + 1)^{-\beta}.$$

The carryover quantity $E_{T,\beta}^{\mathrm{cond}}(b)$ is an endpoint-aware expected replacement-error upper bound. Neither it nor the complete display is asserted to be an exact bias, maximal carryover error, exact estimator risk, or minimax risk.

Proof. **Proof.** Put $r = r_b = b/(b+1)$, $s = 1/(b+1)$, $q_b = r^{-b} = (1+1/b)^b$, and $\lambda = r - s = (b-1)/(b+1)$. The assignment process is the stationary symmetric two-state Markov chain with transition matrix

$$P = \begin{pmatrix} r & s \\ s & r \end{pmatrix}.$$

It is outcome-independent and belongs to $\mathcal{D}_T$. Stationarity and the Markov property give

$$\upsilon_{t,b,a} = \frac{1}{2} r^{k_t} \geq \frac{1}{2} r^b = \frac{1}{2q_b} > \frac{1}{2e} > 0, \tag{1}$$

because $b \log(1 + 1/b) < 1$. Equality in the weak inequality is obtained when $k_t = b$. Since $q_b \to e$, the uniform infimum is $1/(2e)$, approached but not attained at finite b.

Fix $Y \in \mathcal{P}_{T,\beta}(C)$, write $y_{t,a} = Y_t(\mathbf{a}_{1:t})$, and define Z_t and $\widetilde{\theta}_b$ as in the statement. The exposure events for the two arms are disjoint, and inverse-probability weighting yields

$$\mathbb{E} Z_t = y_{t,1} - y_{t,0}, \qquad \mathbb{E}\widetilde{\theta}_b = \mathrm{GATE}_T(Y). \tag{2}$$

Moreover,

$$\mathbb{E} Z_t^2 = 2r^{-k_t}(y_{t,1}^2 + y_{t,0}^2) \leq 4B^2 r^{-k_t}. \tag{3}$$

When $y_{t,0} = y_{t,1} = B$, the mean in (2) is zero and equality holds in the diagonal variance bound.

Let $u = t + h$ with $1 \leq h \leq b$. Opposite-arm exposure events are disjoint, whereas

$$\Pr(H_{t,b,a} = H_{u,b,a} = 1) = \frac{1}{2} r^{k_t + h}.$$

Since $k_u \geq h$, the number $\rho = 2r^{h-k_u}$ satisfies $\rho \geq 2$. Expanding the covariance exactly gives

$$\mathrm{Cov}(Z_t, Z_u) = (\rho - 1)(y_{t,1} y_{u,1} + y_{t,0} y_{u,0}) + y_{t,1} y_{u,0} + y_{t,0} y_{u,1}. \tag{4}$$

Thus

$$|\mathrm{Cov}(Z_t, Z_u)| \leq 2(\rho - 1)B^2 + 2B^2 = 4B^2 r^{h-k_u}, \tag{5}$$

with equality when all four endpoint values equal B.

If $u - t > b$, put $d = u - b - t \geq 1$. Conditional on W_{u-b}, the final b transitions in the exposure window stay at that state with probability r^b, so

$$\mathbb{E}(Z_u \mid W_{u-b} = 1) = 2y_{u,1}, \qquad \mathbb{E}(Z_u \mid W_{u-b} = 0) = -2y_{u,0}.$$

The transition identity

$$P^d(i,j) = \frac{1}{2}\{1 + (2i-1)(2j-1)\lambda^d\}$$

implies

$$|\mathbb{E}(Z_u \mid W_{1:t}) - \mathbb{E} Z_u| \leq 2B\lambda^d.$$

Because Z_t is measurable with respect to $W_{1:t}$ and $\mathbb{E}|Z_t| = |y_{t,1}| + |y_{t,0}| \leq 2B$,

$$|\mathrm{Cov}(Z_t, Z_u)| \leq 4B^2 \lambda^d. \tag{6}$$

Again equality holds for the constant endpoint array. Summing (3), twice the near-pair bounds (5), and twice the far-pair bounds (6) gives

$$\mathrm{Var}\left(\sum_{t=1}^T Z_t\right) \leq B^2 V_{T,b}^{\dagger}. \tag{7}$$

For the admissible constant potential-outcome array $Y_t(\cdot) = B$, equality holds simultaneously in every term used in (7). Hence

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \mathrm{Var}_{\pi^{(b)}}\left(\sum_{t=1}^T Z_t\right) = B^2 V_{T,b}^{\dagger}. \tag{8}$$

This includes the boundary cases: for $T = b = 1$, the variance is $4B^2 = B^2 V^\dagger_{1,1}$, and for $T = 2, b = 1$, direct enumeration gives $V^\dagger_{2,1} = 20$. From (2), Jensen's inequality, and (8),

$$\mathbb{E}|\widetilde{\theta}_b - \mathrm{GATE}_T(Y)| \le \frac{B}{T}\sqrt{V^\dagger_{T,b}}. \tag{9}$$

We next bound the exact variance expression. Since $r^{-k_t} \le q_b$,

$$4\sum_{t=1}^{T} r^{-k_t} \le 4q_b T. \tag{10}$$

Also,

$$\sum_{h=1}^{b}\sum_{t=1}^{T-h} r^{h-k_{t+h}} \le T\sum_{h=1}^{b} r^{h-b} = Tb(q_b - 1), \tag{11}$$

and

$$8\sum_{d=1}^{(T-b-1)_+}(T-b-d)\lambda^d \le 8(T-b)\frac{\lambda}{1-\lambda} = 4(T-b)(b-1). \tag{12}$$

Equations (10)–(12) give the first displayed bound on $V^\dagger_{T,b}$. For $b \ge 3$, the binomial expansion of q_b gives

$$e - q_b \ge \frac{1}{b} - \frac{1}{3b^2}, \qquad q_b \le e < \frac{11}{4},$$

and consequently

$$8b(e - q_b) - 4(q_b - 1) > 0. \tag{13}$$

The difference between $(8e-4)Tb$ and the right side of (10)–(12) is

$$T\{8b(e - q_b) - 4(q_b - 1)\} + 4b(b-1) > 0.$$

For $b = 1$, direct summation gives $V^\dagger_{T,1} = 16T - 12 < (8e-4)T$; for $b = 2$, the preceding bound gives $V^\dagger_{T,2} \le 33T - 8 < (16e - 8)T$. Therefore

$$V^\dagger_{T,b} < (8e-4)Tb. \tag{14}$$

For fixed b, geometric summation in the three exact terms yields

$$\lim_{T\to\infty}\frac{V^\dagger_{T,b}}{Tb} = \frac{4q_b}{b} + 8(q_b - 1) + 4\frac{b-1}{b}.$$

Letting $b \to \infty$ gives $8e - 4$. Together with (14), this proves that $8e - 4$ is the smallest uniform coefficient.

It remains to sharpen the replacement term. For $t \le b+1$, the event $H_{t,b,a} = 1$ fixes the entire history through t at arm a, so replacement error is zero. Now fix indices in the exact domain

$$t \ge b+2, \qquad b+1 \le \ell \le t-1, \qquad a \in \{0,1\}. \tag{15}$$

On $H_{t,b,a} = 1$, one has $W_{t-b} = a$, and the upper bound $\ell \le t-1$ ensures that $W_{t-\ell}$ is a well-defined coordinate of the assignment path. The Markov property makes the past $W_{1:t-b-1}$ conditionally independent of the future segment $W_{t-b+1:t}$ given W_{t-b}. Since the stationary symmetric chain is reversible, its $(\ell - b)$-step backward transition equals its forward transition. Therefore

$$\Pr(W_{t-\ell} \ne a \mid H_{t,b,a} = 1) = \Pr(W_{t-\ell} \ne a \mid W_{t-b} = a) = \frac{1-\lambda^{\ell-b}}{2}. \tag{16}$$

On the same exposure event, boundedness and the historical envelope imply

$$|Y_t(W_{1:t}) - y_{t,a}| \le \min\left\{2B,\ C\sum_{\ell=b+1}^{t-1}(1+\ell)^{-(1+\beta)}\mathbf{1}\{W_{t-\ell} \ne a\}\right\}. \tag{17}$$

The map $x \mapsto \min\{2B, x\}$ is concave and nondecreasing on $[0, \infty)$. Conditioning in (17), applying Jensen's inequality, and using (16) for every index in its exact range gives

$$\mathbb{E}[|Y_t(W_{1:t}) - y_{t,a}| \mid H_{t,b,a} = 1] \le \min\left\{2B, \frac{C}{2} \sum_{\ell=b+1}^{t-1} (1+\ell)^{-(1+\beta)}(1-\lambda^{\ell-b})\right\}. \tag{18}$$

Because $\mathbb{E}[H_{t,b,a}/\upsilon_{t,b,a}] = 1$, the triangle inequality applied to the two arms and (18) proves

$$\mathbb{E}|\widehat{\vartheta}_b - \widetilde{\theta}_b| \le E^{\mathrm{cond}}_{T,\beta}(b). \tag{19}$$

Dropping the cap and interchanging the two finite sums gives

$$E^{\mathrm{cond}}_{T,\beta}(b) \le \frac{C}{T} \sum_{\ell=b+1}^{T-1} (T-\ell)(1+\ell)^{-(1+\beta)}(1-\lambda^{\ell-b}). \tag{20}$$

Since $0 \le 1 - \lambda^{\ell-b} \le 1$, monotonicity and the integral comparison imply

$$\begin{aligned} E^{\mathrm{cond}}_{T,\beta}(b) &\le \frac{C}{T}(T-b-1)_+ \sum_{\ell=b+1}^{\infty} (1+\ell)^{-(1+\beta)} \\ &\le \frac{C}{\beta} \frac{(T-b-1)_+}{T}(b+1)^{-\beta}. \end{aligned} \tag{21}$$

All sums in (19)–(21) are zero when empty, so the formulas cover $T = 1$, $b = 1$, and $b = T$. Finally, combine (9), (14), and (19)–(21) with the triangle inequality. This gives both the finite certificate $U^{\mathrm{cond}}_{T,\beta}(b)$ and its coarse upper bound with coefficient C/β. The exactness in (8) concerns only the auxiliary homogeneous-history variance; none of the replacement or risk inequalities is asserted to be an exact bias, maximal carryover error, exact observed-data risk, or minimax risk. □ □

Justification. The stationary Markov exposure probabilities support an explicit observed-data Horvitz–Thompson estimator; variance is controlled by Markov covariance decay, and carryover is controlled by an expected replacement-error upper bound obtained from $\mathbb{E}[H_{t,b,a}/\upsilon_{t,b,a}] = 1$. *Gap.* The inequality is a worst-case risk upper bound. Its carryover summand is not claimed to equal the estimator's bias or to be minimax sharp. *Consumer.* For any chosen washout scale b, the displayed formula gives a directly evaluable design-based risk guarantee with positive exposure probabilities.

Lemma 9 (lem:common-design-adaptation-bracket)**.** *Fix $C > 0$ and set*

$$C_0^\star = \sqrt{(8e-4)\left(1 + 2^{-1/(2\beta_+ + 1)}\right)}\, B + \frac{C}{\beta_-}.$$

For $b \in \{1, \dots, T\}$, put $r_b = b/(b+1)$, $k_t = \min\{b, t-1\}$, $\lambda_b = (b-1)/(b+1)$,

$$V^\dagger_{T,b} = 4\sum_{t=1}^{T} r_b^{-k_t} + 8\sum_{h=1}^{b}\sum_{t=1}^{T-h} r_b^{h-k_{t+h}} + 8\sum_{d=1}^{(T-b-1)_+} (T-b-d)\lambda_b^d,$$

$$E^{\mathrm{cond}}_{T,\beta_-}(b) = \frac{2}{T}\sum_{t=b+2}^{T} \min\left\{2B, \frac{C}{2}\sum_{\ell=b+1}^{t-1}(1+\ell)^{-(1+\beta_-)}(1-\lambda_b^{\ell-b})\right\}, \qquad U^{\mathrm{cond}}_{T,\beta_-}(b) = \frac{B}{T}\sqrt{V^\dagger_{T,b}} + E^{\mathrm{cond}}_{T,\beta_-}(b),$$

with empty inner sums equal to zero. Put $b_- = \lceil T^{1/(2\beta_- + 1)} \rceil$ and define

$$N^{\mathrm{fin}}_T = \begin{cases} B, & T = 1, \\ U^{\mathrm{cond}}_{T,\beta_-}(b_-), & T \ge 2, \end{cases} \qquad N^\star_T = \begin{cases} B, & T = 1, \\ \min\left\{B, \min_{1 \le b \le T} U^{\mathrm{cond}}_{T,\beta_-}(b)\right\}, & T \ge 2, \end{cases}$$

and

$$D_T^{\mathrm{bin}} = B2^{-(T-1)}\binom{T-1}{\lfloor (T-1)/2\rfloor}.$$

Then, for every $T \geq 1$,

$$1 \leq A_T(C) \leq \frac{N_T^\star}{D_T^{\mathrm{bin}}} \leq \frac{N_T^{\mathrm{fin}}}{D_T^{\mathrm{bin}}} \leq \frac{\sqrt{2}}{B}\left[\sqrt{(8e-4)\left(1+2^{-1/(2\beta_-+1)}\right)}\,B + \frac{C}{\beta_-}\right] T^{1/(4\beta_-+2)}.$$

At $T = 1$,

$$A_1(C) = 1, \qquad N_1^\star = N_1^{\mathrm{fin}} = D_1^{\mathrm{bin}} = B.$$

This equality is attained by the common fixed-before-assignment rule $W_1 \sim \mathrm{Bernoulli}(1/2)$ with $\widehat{\theta}_1 = (2W_1 - 1)Y_1^{\mathrm{obs}}$. For $T \geq 2$, the upper certificate is attained by one rule selected before assignment using only T, B, C, β_-: compare the persistent homogeneous-path fair-coin procedure

$$\widehat{\theta}^{\mathrm{hom}} = (2R-1)T^{-1}\sum_{t=1}^{T} Y_t^{\mathrm{obs}}, \qquad W_t = R, \qquad R \sim \mathrm{Bernoulli}(1/2),$$

with the stationary Markov switchback and observed-data Horvitz–Thompson estimator at the smallest scale minimizing $U_{T,\beta_-}^{\mathrm{cond}}(b)$. For $\mu_a = T^{-1}\sum_{t=1}^{T} Y_t(\mathbf{a}_{1:t})$, the persistent branch has risk $(|\mu_0| + |\mu_1|)/2 \leq B$ and exact worst-case risk B, attained by the constant array $Y_t(\cdot) = B$. This common rule is independent of the realized β. In its Markov branch, $\upsilon_{t,b,a} \geq 1/(2q_b) > 1/(2e)$, where $q_b = (1 + 1/b)^b$, with unattained uniform infimum $1/(2e)$. The final coefficient is endpoint-specific because β_- is known to the common design; it is not replaced by the larger β_+-uniform constant $C_0^\star$. The denominators are the exact binomial Bayes lower bound on oracle risk, not the unknown exact oracle risk except at $T = 1$. This is a common-design polynomial upper certificate; it proves neither K1 divergence, K2 subpolynomial adaptation, nor sharpness.

Proof. **Proof.** For every $\beta \in [\beta_-, \beta_+]$ and $\ell \geq 1$,

$$C(1+\ell)^{-(1+\beta)} \leq C(1+\ell)^{-(1+\beta_-)}.$$

The defining envelope and the common boundedness condition therefore give

$$\mathcal{P}_{T,\beta}(C) \subseteq \mathcal{P}_{T,\beta_-}(C). \tag{1}$$

Construct the persistent homogeneous-path procedure by drawing $R \sim \mathrm{Bernoulli}(1/2)$ before outcomes, setting $W_t = R$ for every t, and using

$$\widehat{\theta}^{\mathrm{hom}} = (2R-1)\frac{1}{T}\sum_{t=1}^{T} Y_t^{\mathrm{obs}}. \tag{2}$$

For a fixed array write

$$\mu_a = \frac{1}{T}\sum_{t=1}^{T} Y_t(\mathbf{a}_{1:t}), \qquad a \in \{0,1\}.$$

Then $\mathrm{GATE}_T(Y) = \mu_1 - \mu_0$. Conditional on $R = 1$, the error in (2) is $|\mu_0|$; conditional on $R = 0$, it is $|\mu_1|$. Hence

$$\mathbb{E}\left|\widehat{\theta}^{\mathrm{hom}} - \mathrm{GATE}_T(Y)\right| = \frac{|\mu_0| + |\mu_1|}{2} \leq B. \tag{3}$$

The admissible constant array $Y_t(\cdot) = B$ attains equality, so the branch has exact worst-case risk B. It is fixed before assignment, outcome-independent, and independent of β; no positivity claim is made for it.

Let $T \geq 2$. Compare the persistent procedure with the finitely many Markov procedures of Lemma $\mathrm{lem:markov-washout-oracle-upper}$, all evaluated at β_-, and select before assignment a procedure whose certificate equals

$$N_T^\star = \min\left\{B, \min_{1\leq b\leq T} U_{T,\beta_-}^{\mathrm{cond}}(b)\right\}. \tag{4}$$

If the selected branch is Markov, use the smallest minimizing scale. By (1), this one procedure, which depends only on T, B, C, β_-, satisfies

$$\sup_{Y \in \mathcal{P}_{T,\beta}(C)} \mathbb{E}_\pi \left| \widehat{\theta} - \mathrm{GATE}_T(Y) \right| \leq N_T^\star \quad \text{for every } \beta \in [\beta_-, \beta_+]. \tag{5}$$

Lemma lem : exact − positive − oracle − risk gives

$$R_T(\beta; C) \geq D_T^{\mathrm{bin}} > 0 \quad \text{for every } \beta \in [\beta_-, \beta_+]. \tag{6}$$

Divide (5) by (6), take the supremum over β, and use the selected procedure as a candidate in the infimum defining $A_T(C)$. This proves

$$A_T(C) \leq \frac{N_T^\star}{D_T^{\mathrm{bin}}}. \tag{7}$$

Conversely, for any fixed design and estimator, its worst-case risk at each β is at least the infimum $R_T(\beta; C)$. Division is valid by (6), so its ratio is at least one for every β. Taking the supremum and then the infimum proves $A_T(C) \geq 1$.

The admissibility of $b_- = \lceil T^{1/(2\beta_-+1)} \rceil$ and the definitions imply

$$N_T^\star \leq U_{T,\beta_-}^{\mathrm{cond}}(b_-) = N_T^{\mathrm{fin}}. \tag{8}$$

Theorem thm : oracle − rate, evaluated at β_-, gives

$$N_T^{\mathrm{fin}} \leq \left[\sqrt{(8e-4)\left(1 + 2^{-1/(2\beta_-+1)}\right)}\, B + \frac{C}{\beta_-} \right] T^{-\beta_-/(2\beta_-+1)}. \tag{9}$$

Also, (6) gives $D_T^{\mathrm{bin}} \geq B/\sqrt{2T}$. Dividing (9) by this lower bound and using

$$\frac{1}{2} - \frac{\beta_-}{2\beta_- + 1} = \frac{1}{4\beta_- + 2}$$

proves the final polynomial inequality.

For $T = 1$, Theorem thm : oracle − rate gives $R_1(\beta; C) = B$, while (3) and its constant-array witness give exact worst-case risk B for the same common procedure. Its ratio is therefore one for every β, and the general lower-bound argument gives $A_1(C) = 1$. The definitions give $N_1^\star = N_1^{\mathrm{fin}} = D_1^{\mathrm{bin}} = B$; the last coefficient inequality at $T = 1$ follows from Theorem thm : oracle − rate.

Any selected Markov branch inherits the exposure floor $v_{t,b,a} \geq 1/(2q_b) > 1/(2e)$ and its unattained uniform infimum from Lemma lem : markov − washout − oracle − upper. That statement does not apply to the persistent branch. The numerator coefficient is endpoint-specific because the common procedure may use the known β_-. The denominator in (7) is only the exact binomial Bayes lower bound on oracle risk, except at $T = 1$; consequently the result remains a polynomial certificate and proves neither K1, K2, nor sharpness. □ □

Justification. A single Markov switchback tuned to the largest class $\mathcal{P}_{T,\beta_-}(C)$, combined with nestedness and the exact oracle-risk floor, yields an explicit simultaneous oracle ratio. *Gap.* The certified factor is polynomial, so it proves neither K1 divergence nor K2 subpolynomial adaptation and is not asserted to be sharp. *Consumer.* It supplies the strongest proved upper control on the value of knowing the decay index before randomization.

7 Interpretation

When β is known, compute $U_{T,\beta}^{\mathrm{cond}}(b)$ over all integer scales and compare its minimum with B. Select before assignment either the persistent homogeneous-path fair-coin procedure or the smallest minimizing Markov washout scale. The persistent branch has exact worst-case risk B; the Markov variance is exact only for the auxiliary homogeneous-history process, and its carryover term is an expected replacement-error upper bound. The scale $b_\beta = \lceil T^{1/(2\beta+1)} \rceil$ gives the displayed power-rate certificate. At $T = 1$, the persistent rule reduces to the fair-Bernoulli procedure and yields $R_1(\beta; C) = B$ and $A_1(C) = 1$. Without knowing β, tune the Markov candidate to β_- and compare it with the same persistent branch; this controls every nested class but yields only the displayed polynomial oracle ratio.

Technical internal limitation (diagnostic only)

The exact B risk is exact only for the persistent selector branch; it is not the oracle risk except at $T = 1$. The identity $B^2 V^{\dagger}_{T,b}$ concerns the auxiliary homogeneous-history variance, not the observed-data estimator's exact risk. The conditional quantity $E^{\mathrm{cond}}_{T,\beta}(b)$ retains its internal per-time cap $2B$ and is a capped expected replacement-error upper bound, not exact bias or maximal carryover error. The exact binomial expression is an oracle-risk lower bound, not oracle risk itself except at $T = 1$. The beta-aware upper exponent remains unmatched by the universal $1/2$ lower exponent. Markov positivity applies only to the Markov branches. The B-elbow improvement supplies neither the missing two-class complementarity inequality nor a decay-agnostic $T^{o(1)}$ multiscale selector.

8 Honest scope

The paper proves computable finite design-risk upper bounds, the exact B-risk persistent homogeneous-path fair-coin branch, the exact worst-case auxiliary homogeneous-history variance for the Markov construction, the sharp uniform auxiliary-variance coefficient $8e - 4$, the Markov-family exposure floor $1/(2e)$, the exact conditional pre-window mismatch probability, capped endpoint-aware conditional replacement control, the beta-aware ceiling rate, exact identities $R_1(\beta; C) = B$ and $A_1(C) = 1$, finite selectors with a B elbow, an oracle bracket with unmatched exponents, and an endpoint-specific common-design polynomial certificate. It does not prove a matching growing-horizon oracle rate, divergent adaptation cost, a subpolynomial common procedure, a sharp order of $A_T(C)$, or that adaptation is never free. K1 and K2 remain jointly open and logically compatible.

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